

Bovaer + Rumensin Greenhouse Gas Emissions Reduction Protocol

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1 Introduction

The Bovaer & Rumensin Greenhouse Gas (GHG) Emissions Reduction Protocol was developed by Elanco to provide a standardized methodology for quantifying GHG emissions reductions. This protocol specifically addresses reductions associated with the joint use of Bovaer (3-nitrooxypropanol or 3-NOP1) and Rumensin in U.S. dairy cattle.

Bovaer is a feed ingredient designed to reduce methane for lactating dairy cattle. Rumensin, an ionophore, is approved for improving milk production efficiency in lactating cattle and for increasing weight gain and preventing coccidiosis in growing cattle. This comprehensive protocol references the Bovaer Feed Ingredient Protocol and represents a new and standalone Protocol. It aims to holistically account for on-farm GHG emissions reductions by considering:

- The enteric methane reduction effect of Bovaer in lactating dairy cattle.
- The enteric effects of Rumensin in dry cattle and replacement heifers.
- The manure and feed impacts of Rumensin across lactating cattle, dry cattle, and replacement heifers.

Livestock supply chains contribute significantly to global greenhouse gas emissions, accounting for 14.5% of anthropogenic GHGs (Gerber et al., 2013). Within this, enteric methane is a major concern, representing 43% of GHG emissions from U.S. dairy farms (Rotz et al., 2021). Given that methane has a global warming potential 27 times greater than CO₂ over a 100-year horizon (IPCC, 2021), strategies to reduce enteric methane are crucial for achieving supply chain sustainability goals.

As a global leader in animal health, Elanco is dedicated to innovating products and services that prevent and treat disease in farm animals, thereby creating value for farmers, stakeholders, and society. Elanco developed this specific methodology to provide clear guidance on the credible quantification and monitoring of GHG reductions when Bovaer and Rumensin are used within a single farm operation.

It is important to note that operations implementing only one of these feed ingredients should refer to their respective independent methodologies (the Bovaer Feed Ingredient Protocol or the Rumensin Dairy Cattle Feed Additive Protocol). The reductions quantified by following this combined methodology are intended to be claimed as Scope 3 reductions within the dairy supply chain. This document outlines a comprehensive approach to quantify these reductions in a complete, consistent, transparent, and accurate manner.

2 Project Definition

In this project, the intervention activities include feeding Bovaer and Rumensin in lactating cattle, and Rumensin in heifers and dry cows. These feed ingredients each have the effect of reducing GHG emissions associated with the production of dairy products. Dairy producers purchase feed ingredients from their feed mills and feed them as part of their existing diets formulated by nutritionists, typically fed as part of a mineral mix or through a micromachine.

The lactating dairy cattle included in the project must consume both Rumensin and Bovaer, and any involved dry cattle or heifers must consume Rumensin. The replacement animals included in this project must be those animals required to maintain the lactating herd at the

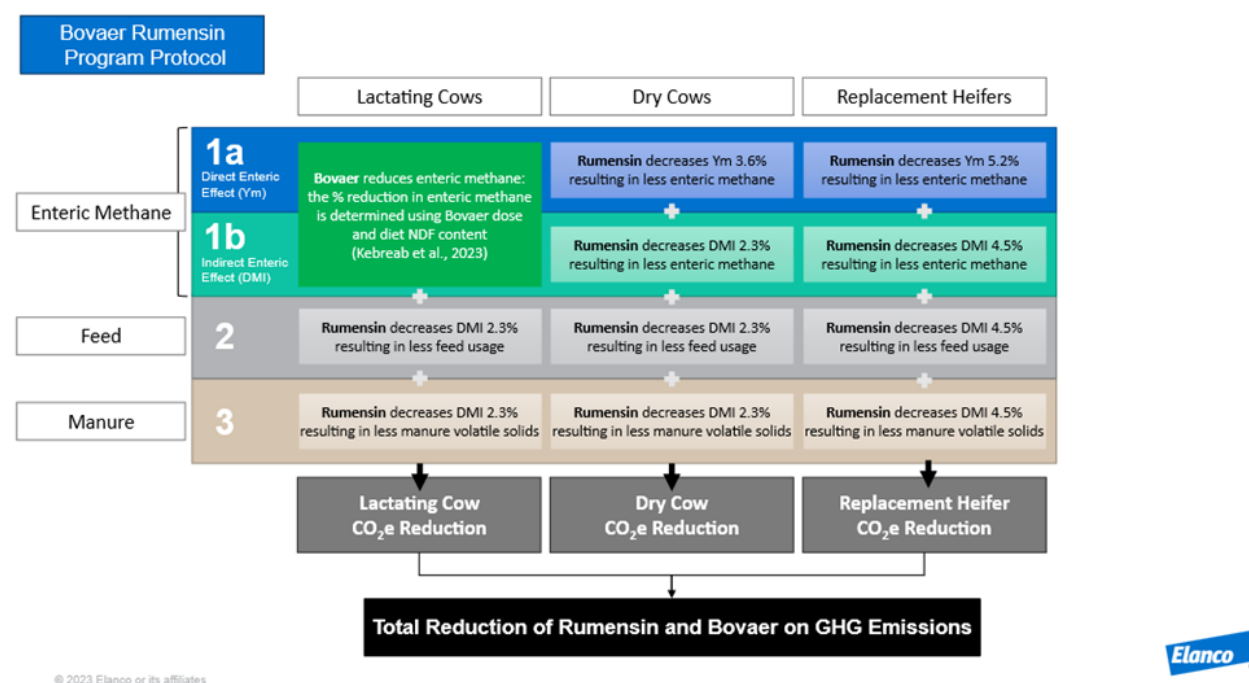
participating operation; that is, heifers raised for other operations are ineligible. These cattle do not need to be collocated.

Figure 2.1 shows an overview of the contribution of each ingredient to each animal type and emissions source.

Table 2.1 Eligible Animal Types, Definitions, and Eligible Boundaries

Animal Type	Definition	Feed	Enteric	Manure
Lactating cows	A dairy cow producing milk	Rumensin	Bovaer	Rumensin
Dry cows	An adult, non-lactating dairy animal; may include replacement heifers in late stage of pregnancy grouped with mature non-lactating cows	Rumensin	Rumensin	Rumensin
Replacement heifers	Growing female dairy animals intended for future milk production, consuming a partial mixed ration or total mixed ration	Rumensin	Rumensin	Rumensin

Figure 2.1 Quantified Boundaries; summary impact of feeding Rumensin and Bovaer in a single operation



2.1 Impact on Yield

Neither Rumensin nor Bovaer have a negative effect on fat and protein corrected milk (FPCM). Current data suggests there is no impact of Bovaer on fat and protein corrected milk yields (Jayanegara et al., 2018; Kim et al., 2020; Hristov et al., 2022; Martins et al., 2025;

Pupo et al., 2025). Therefore, no impact to component-corrected milk yield at the consumer product good level is expected with the feeding of Bovaer. The anticipated volume of fat and protein corrected milk for a pilot enrollment is approximately 22,194,048.85 kg. This is based on a 2024 average animal production of 24,178 lbs per cow (September 2025 NASS Milk Production Report), with 4.18% butterfat and 3.36% crude milk protein (CA Annual Report, chosen to match the intended supply shed geography).

Rumensin is approved for the improvement of milk production efficiency (production of marketable solids-corrected milk per unit of feed intake) in dairy cows. Rumensin works by shifting rumen bacterial populations to produce more propionate which is a more efficient use of feedstuffs because there is a reduction in the amount of energy lost as carbon dioxide and methane. A meta-analysis on the impact of Rumensin on productivity outcomes in lactating dairy cattle was conducted using 71 trials containing data from 255 trial sites and 9,677 cows examining milk production and composition. A wide range of doses encompassing the 11-22 g/ton DM dose requirement outlined in this protocol) were represented. Studies lacking a control group, randomization, or reporting of measures of dispersion or P-values were excluded from consideration in the analysis. Rumensin use in dry and lactating dairy cattle significantly decreased dry matter intake (DMI) by 2.3% and increased milk yield by 2.3%. Because significant heterogeneity was observed with milk components, the reported yield and component percentages were used to calculate solids-corrected milk (SCM) differences between control and Rumensin treatments, resulting in a 0.5% increase in SCM with Rumensin. Therefore, Rumensin-adjusted milk and component yields are reflected in baseline nitrogen retention calculations within this protocol. Rumensin increases milk production efficiency through a combination of increased SCM and decreased feed required, therefore there are no concerns for impact to total milk production and subsequent leakage. Monthly milk production data collected as part of this program.

2.2 Causality

Bovaer directly addresses enteric methane by preventing its formation in the rumen through the targeted inhibition of the final reaction in Archaeal methane production. In May 2024, the U.S. Food and Drug Administration (FDA) completed its review, confirming Bovaer meets the requirements for marketing and sale for use in U.S. lactating dairy cattle.

Peer-reviewed literature demonstrates that Bovaer can reduce enteric methane emissions by approximately 30%, or 1.2 metric tons (MT) per cow of CO₂e over a 305-day lactation, depending on dose and diet. These reductions can be further refined through modeling for more accurate estimates (Kebreab et al., 2023). Models suggest that widespread implementation of Bovaer could lead to an estimated net reduction of around 13.9 billion kg CO₂e in GHG emissions for the U.S. dairy industry (Uddin et al., 2022).

Many dairy operations that feed Bovaer to lactating cattle also utilize Rumensin for their lactating and dry cattle, as well as replacement heifers. This combined usage highlights the need for a comprehensive quantification protocol.

Rumensin (monensin) is an ionophore that enhances propionate production, leading to improved milk production efficiency. This improvement typically ranges from 2% to 4% when fed at concentrations of 11 to 22 g/ton (NADA 095-735). For growing cattle, including replacement heifers, Rumensin increases the rate of weight gain and helps prevent and control coccidiosis.

Beyond milk production efficiency, Rumensin contributes to GHG emissions reductions. By increasing propionate production and hydrogen utilization, it creates a hydrogen sink that competes with methanogenesis, resulting in a 3.6% decrease in the proportion of gross energy intake converted into methane and a 2.3% decrease in dry matter intake. These effects lead to overall farm absolute annual GHG reductions of approximately 0.5 MT CO_{2e} per lactating animal through both direct enteric methane reduction and indirect impacts on manure and feed production. The indirect reduction effect on farm feed and manure GHG emissions stems from Rumensin's ability to decrease dry matter intake while maintaining or increasing productivity.

Research indicates that Rumensin and Bovaer are effective at decreasing methane independently, whether fed separately or in combination (Romero-Perez et al., 2016; Vyas et al., 2018). Given their distinct and non-competing modes of action, the enteric methane reduction effects of each feed ingredient are considered independent.

Crucially, under this protocol, the enteric methane effect of Rumensin is not accounted for in lactating cattle that are also consuming Bovaer. This ensures accurate and non-overlapping quantification. All quantification of outcomes for both feed ingredients and across all cattle groups is based on established IPCC methodologies and robust peer-reviewed data and models.

Bovaer reduces enteric methane and thus absolute emissions associated with the production of consumer goods in the dairy supply chain. Feeding Bovaer does not have direct benefit to dairy producers and as such requires investment and engagement from supply chain actors who seek to reduce their Scope 3 Category 1 emissions. Bovaer reduces methane emissions in lactating dairy cattle by approximately 30% each day it is fed (Jayanegara et al., 2018; Dijkstra et al., 2018; Kim et al., 2020; Hristov et al., 2022; Kebreab et al., 2023). Emissions reductions associated with feeding Bovaer are permanent and cannot be reversed.

Causality for dairy processors, consumer packaged goods companies, and retailers is thus achieved through a dual mechanism:

- **Financing:** Supply chain entities involved in the projects invest directly in the implementation of Bovaer. Producers are effectively reimbursed for the cost of the product and receive additional incentive for engaging in Bovaer programs and sharing their data in a way that enables credible claiming throughout the value chain.
- **Education and Training:** Elanco Animal Health, nutritionists, and staff from dairy processors engage—with the support of the financing partners—in ongoing education and training to producers on the proper use of Bovaer, the impact of Bovaer on supply chain emissions, and the connection between sustainability programs and long-term market access.

Prior to 2024, Bovaer was not available for use in the United States and therefore not fed. All Bovaer use in the United States today is implemented through supply chain collaboration as part of emissions reductions programs and would not otherwise occur. Globally and within the United States, Bovaer market penetration is less than 5%.

Daily Rumensin use required in this program is driven by Bovaer use. While Rumensin has been commercially available for almost 20 years, pressures within the industry make choosing to feed Rumensin a daily decision. Although high feed costs effects on feed efficiency has greater returns during times of high feed cost, producers and nutritionists frequently remove it from the diet during these economically burdening times. This paradox is

due to the added up-front cost to the producer and difficulty measuring farm-level return on investment to the operation due to high variability and insufficient quality of farm-level data. Variability on farms and multiple potential outcomes for the increased dietary energy gained from Rumensin (including growth, productivity, maintenance, tissue gain, and/or reduction in dry matter intake) make effects of an efficiency enhancing ingredient difficult to measure on-farm. Participating in this program provides an additional benefit through financial support from the supply chain for the continued use of Rumensin which drives more consistent Rumensin use and ensures use is on-label and at sufficient dose to see the benefits described herein.

Additionally, it is increasingly important that all ingredients which reduce enteric methane are considered, as many producers may be incentivized to remove Rumensin from the diet to participate in carbon programs associated with different or new enteric methane reduction technologies. These replacements carry a substantial risk of mitigating less GHG emissions than are occurring with currently used technologies like Rumensin. Removal of Rumensin in these cases represents a leakage scenario that may not be accounted for in those programs.

Claiming entities financing the feed ingredients in this program invest in a holistic approach to enteric methane mitigation and therefore cause all reductions described herein.

Emissions reductions associated with feeding Bovaer and Rumensin are permanent and cannot be reversed. This creates a significant opportunity to make progress in GHG emissions reductions via a financial incentive from the supply chain. Participating in a protocol and carbon market provides an incentive through financial support from inset or offset markets to support the long-term use of Bovaer. Eligibility and use of this program creates a data stream the supply chain will need for credibly delivering on greenhouse gas reduction commitments.

3 Eligibility

The operations that affect the SSRs described in 4.1.1 are specifically the feeding of both Bovaer and Rumensin to lactating animals, and optionally the feeding of Rumensin to dry cows and heifers. Adjustments to the ration may occur due to the feeding of Rumensin as well.

Two sets of eligibility criteria are outlined below – one pertaining to the feeding of Bovaer, one to Rumensin, and one governing all participation. To be eligible for quantification of reductions from either intervention, all eligibility criteria must be met for the cattle types participating in the project.

Table 3.1 *Bovaer in Lactating Dairy Cattle Eligibility Requirements*

Eligibility Criteria Summary	
Formulated Dose	The formulated dose, weighted average across all lactating dairy cattle involved in the project, must be on-label (As of May 2024: 60-80 mg 3-NOP per kg total ration dry matter (DM), fresh feed only) ¹ .
Actual Dose	If the actual delivered dose is not on-label but within scientific range (37-80 ppm), alternate documentation is required. If the actual dose is on-label, no further documentation is required.

¹ In the event of a label change that changes the allowable dose range, evidence of proper weighted-average formulation shall conform to the most current label. In case of conflict between the label and this document, the label shall prevail.

Animal Types	Lactating dairy cattle
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Table 3.2 *Rumensin in Lactating Dairy Cattle, Dry Dairy Cattle, and Replacement Heifers Eligibility Requirements*

Eligibility Criteria Summary	
Formulated Dose	The formulated dose, weighted average across all cattle involved in the project must be on-label. Label instructions are highlighted below ¹ .
Actual Dose	If the actual delivered dose is not on-label but within scientific range (10 - 32 g/ton), alternate documentation is required. If the actual dose is on-label, no further documentation is required.
Animal Types	Lactating dairy cattle required; Dry Cows or Replacement Heifers (both optional). Replacement Heifers must be consuming a partial mixed ration (PMR) or a total mixed ration (TMR) and the feed data must be available on all included animals. Milk replacer may not be a part of the PMR.

Label instructions for Rumensin include the following doses by feeding system and cattle type:

- Dairy cattle (dry and lactating cows):
 - Total Mixed Rations ("complete feed"): 11 to 22 g/ton monensin on a 100% dry matter basis
 - Component Feeding Systems (including top dress):
 - 185 to 660 mg/head/day monensin to lactating cows
 - 115 to 410 mg/head/day monensin to dry cows
 - This provides cows with similar amounts of monensin they would receive by consuming total mixed rations containing 11 to 22 g/ton monensin on a 100% dry matter basis.
- Replacement heifers:
 - 50 to 200 mg per head per day

Diet formulation according to label is an eligibility requirement. These doses are in alignment with the FDA-approved labeling for both Rumensin and Bovaer.

3.1 Voluntary Compliance & Performance Standard

All projects are subject to a legal requirement test to ensure that the GHG reductions achieved by this intervention are not required by federal, state, or local laws or regulations (e.g., air, water quality, water discharge, safety, labor, endangered species protection), or other legally binding mandates. The legal requirement test is applied to each project enrolled in the program. Therefore, if interventions at one project become legally required, it does not affect the other projects in the program.

To satisfy the legal requirement test, each producer whose dairy operation is a project within the program must sign an attestation of voluntary compliance. Attestations must be signed prior to the commencement of verification activities each time the intervention is verified. In

addition, the Monitoring Plan must include procedures that the producer will follow to review existing legal requirements for the intervention location and ascertain and demonstrate that the project passes the legal requirement test.

3.2 Project Start Date

The start date of this intervention is the first date that Bovaer is incorporated into the lactating dairy cattle ration concurrently with Rumensin. This start date may not take place before January 1, 2026.

3.3 Reporting Period

The reporting period is the period of time during which the intervention was implemented. Impact units or reductions for each project can be quantified as frequently as monthly (calendar month).

3.4 Location

Only projects (farms) located in the U.S., or on U.S. tribal lands, are eligible to generate reduction credits under this program.

4 GHG Assessment Boundary

Project boundary for sources, sinks, and reservoirs was determined based on VM0041 methodology for determining project boundaries. More specifically: The spatial extent of the project boundary encompasses all geographic locations of ingredient production, ingredient transport, and project activity locations where feed ingredient is part of the livestock production operation. No sinks are included in this program, as it solely focuses on reduction, not removals.

Quantification of the baseline and project emissions occur separately for Bovaer and Rumensin, and only non-overlapping boundaries are included in this project. Consequently, the assessment boundaries are defined individually.

The process for exclusion of SSRs was to consider the mode of action of each feed ingredient. All GHG sources that are affected by the feed ingredients are included. GHGs that are excluded are not affected by the feed ingredient or may represent a slight further decrease in GHG should they be included, hence driving a conservative approach. Neither Rumensin nor Bovaer alter total enteric carbon dioxide output, and the flux remains constant between baseline and project. In the case of manure carbon dioxide, there is not data available to enable determination of magnitude of effect. Manure carbon dioxide was thus excluded for the purposes of conservativeness; decrease in volume of manure as a consequence of lower dry matter intake likely results in less carbon dioxide emissions due to decomposition of solids. For both manure CO₂ and enteric N₂O, they were excluded due to their exclusion from IPCC guidelines for quantification of manure and enteric emissions. Monitoring is not required for excluded SSRs.

4.1 Bovaer GHG Assessment Boundary

Table 4.1 provides a comprehensive list of the GHG SSRs that may be affected by the project (implementation of Bovaer in lactating dairy cattle) and indicates which SSRs must be included in the GHG Assessment Boundary.

Table 4.1 Description of all sources, sinks, and reservoirs impacted by Bovaer use in Lactating Dairy Cattle under the protocol

Source		Gas	Included	Justification/Explanation
Baseline	Enteric Fermentation	CO ₂	No	No changes in biogenic CO ₂ emissions are expected due to Bovaer implementation (Kim et al., 2020, Appendix 24). CO ₂ emissions are expected to be constant in baseline and project scenario and are not affected by the intervention.
		CH ₄	Yes	CH ₄ emissions from enteric fermentation in the absence of Bovaer implementation represent the major source of emissions in the baseline scenario (Kebreab et al., 2023)
		N ₂ O	No	Enteric N ₂ O emissions are considered de-minimus and are excluded from IPCC and IDF guidelines.
Project	Enteric Fermentation	CO ₂	No	No changes in biogenic CO ₂ emissions are expected due to Bovaer implementation (Kim et al., 2020, Appendix 24). CO ₂ emissions are expected to be constant in baseline and project scenario and are not affected by the intervention.
		CH ₄	Yes	CH ₄ emissions from enteric fermentation are the major source of emissions in the project scenario (Kebreab et al., 2023)
		N ₂ O	No	Enteric N ₂ O emissions are considered de-minimus and are excluded from IPCC and IDF guidelines.[GU2]
	Ingredient Production and Transport	CO ₂	Yes	Emissions from the manufacturing and transport of the feed ingredient intervention are included in this program.
		CH ₄	Yes	Emissions from the manufacturing and transport of the feed ingredient intervention are included in this program.
		N ₂ O	Yes	Emissions from the manufacturing and transport of the feed ingredient intervention are included in this program.
	Manure Decomposition	CO ₂	No	No changes in manure CO ₂ are expected due to Bovaer implementation Nkemka et al., 2019; Owens et al., 2020, 2021
		CH ₄	No	No changes in manure CH ₄ are expected due to Bovaer implementation Nkemka et al., 2019; Owens et al., 2020, 2021
		N ₂ O	No	No changes in manure N ₂ O are expected due to Bovaer implementation Nkemka et al., 2019; Owens et al., 2020, 2021

Bovaer is metabolized by its own mode of action (Duin et al., 201, Appendix 19) and is not practically excreted in the environment (Bampidis et al., 2021). Studies between control and 3-NOP-fed animals have shown no differences in fecal characteristics (volatile solids, carbon, nitrogen or total solids) in forage- or grain-based diets (Nkemka et al., 2019), no differences in manure GHG and NH₃ emissions or manure application to crop yield (Owens et al., 2020, 2021), and no impact on anaerobic digestion (Nkemka et al., 2019). Therefore, because Bovaer does not affect any GHG emission sources outside the identified GHG boundaries, leakage is not relevant to this project and no deductions will be applied to outcomes generated according to this protocol. This protocol assumes no effect and thus is conservative relative to a scenario in which there could be a decrease in dry matter intake.

4.2 Rumensin GHG Assessment Boundary

The GHG Assessment Boundary encompasses the GHG SSRs that may be significantly impacted by project activities. This includes CH₄ emissions from enteric fermentation; CH₄ and nitrous oxide (N₂O) from manure management; and CO₂ and N₂O emissions from feed production.

Table 4.2 Description of all sources, sinks, and reservoirs impacted by Rumensin use under the protocol

Source		Gas	Included?	Justification/Explanation
Baseline and Project	Feed Cultivation	CO ₂	Yes	Emissions from transportation, production, and harvesting cattle feed are reduced by the practices included in this protocol.
		CH ₄	Yes	
		N ₂ O	Yes	
	Enteric Fermentation	CO ₂	No	No changes in biogenic CO ₂ emissions are expected due to Rumensin implementation. CO ₂ emissions are expected to be constant in baseline and project scenarios and are not affected by the intervention.
		CH ₄	Yes, for dry cattle and replacement heifers	Emissions from enteric fermentation in cattle are reduced by the practices included in this protocol but are only eligible for quantification in non-lactating animals.
		N ₂ O	No	Enteric N ₂ O emissions are considered de-minimus and are excluded from IPCC and IDF guidelines.
	Manure Management	CO ₂	No	Manure CO ₂ emissions are considered de-minimus and are excluded from IPCC and IDF guidelines for manure emissions.
		CH ₄	Yes	Emissions from the management of manure are reduced by the practices included in this protocol.
		N ₂ O	Yes	Emissions from the management of manure are reduced by the practices included in this protocol.

Rumensin's mode of action for milk production efficiency results in reduced methane production. The impacts of feeding Rumensin to cattle include both a direct reduction in CH₄ emissions from enteric fermentation and an indirect reduction in enteric fermentation emissions through decreased DMI with no negative effects on health and a positive benefit to milk production efficiency. The persistent benefit of feeding cattle Rumensin is supported by multiple scientific studies that have measured reductions in methane (CH₄) emissions from enteric fermentation. A summary of Rumensin direct methane reduction impacts utilized within this protocol to quantify GHG reductions is highlighted in **Table 4.3**.

These values were calculated by dividing the Rumensin effect on dietary gross energy lost as CH₄ in Model I of Table 4 in Appuhamy et al. for each respective cattle type (-0.23 for lactating and dry cows and -0.33 for replacement heifers and beef) by the average dietary gross energy lost as CH₄ for both dairy cows and beef steers (6.35%; Ym) in Table 3 in the Appuhamy paper.

Due to Rumensin's effect on decreased DMI, the GHG impact of feed cultivation and manure is also reduced, providing an additional GHG benefit. These impacts are quantified through the process and calculations in this protocol for lactating cattle, dry cattle, and replacement heifers.

Figure 2.1 summarizes the eligible boundaries and effects. Thus, this program quantifies the SSRs that are affected directly by the mode of action of feed ingredients and excludes those that are not affected or are assumed constant in the baseline and project scenario (i.e. enteric CO₂).

In the case of manure CO₂, the Rumensin mode of action is further considered. Decrease in manure methane and nitrous oxide emissions are quantified as a result of decreased dry matter intake which decreases produced volatile solids. While those solids will decompose and produce manure carbon dioxide, it is a small amount relative the methane and nitrous oxide emissions. Because neither feed ingredient affects manure carbon dioxide directly, it is complete and conservative to exclude it because lower volume of manure also means lower manure carbon dioxide due to decomposition.

For both manure CO₂ and enteric N₂O, they were excluded due to their exclusion from IPCC guidelines for quantification of manure and enteric emissions.

Table 4.3 Animal Type, Dose Range, and Ym and DMI Reductions Included in Protocol

Animal Type	Quantified in this Protocol?	Literature Dose Range	Ym Reduction	DMI Reduction
Lactating cattle	No, refer to Rumensin Dairy Cattle Feed Additive Protocol	10 – 32 g/ton	3.6% ¹	2.3% ²
Dry cattle	Yes	10 – 32 g/ton	3.6% ¹	2.3% ²
Replacement heifers	Yes	25 – 36 g/ton	5.2% ¹	4.30% ³
¹ Appuhamy et al., 2013 ² Duffield et al. 2008 ³ Gadberry et al., 2022				

4.3 Relevant Sinks, Sources, and Reservoirs

The relevant SSRs to Bovaer activity are enteric fermentation for lactating cows and product manufacturing and transport. The relevant SSRs to Rumensin activity are enteric fermentation for dry cows and heifers, feed cultivation, and manure management.

5 GHG Quantification

Reductions are quantified separately by intervention type and aggregated at the farm level in reporting. In both cases, project emissions are calculated according to available studies and follow IPCC tier-2 aligned approaches. A counterfactual baseline is used in both cases, both of which represent the business-as-usual scenario. In the case of Bovaer, the feed ingredient is not fed in absence of this program and would not otherwise happen. In the case of Rumensin, Bovaer reductions programs drive more consistent usage (across time (days), volume (dose), and herds) of the feed ingredient, and this program offers more holistic accounting of the impacts of the products. All quantification uses primary data submitted by producers through the UpLook tool.

Consequently, emissions reductions are calculated by subtracting project emissions from baseline emissions, as follows:

$$ER_{Total} = ER_{Bovaer} + ER_{Rumensin} \quad (Equation 5.1)$$

Where:

ER_{Total}	=	Total GHG emission reductions from all animal groups during the reporting period (MT CO ₂ e)
ER_{Bovaer}	=	Total emission reductions quantified from lactating dairy cattle fed Bovaer during the reporting period (MT CO ₂ e)
$ER_{Rumensin}$	=	Total emission reductions quantified from lactating dairy cattle, dry cows, and/or replacement heifers fed Rumensin during the reporting period (MT CO ₂ e)

$$ER_{Bovaer} = BE_{Enteric} - PE_{Enteric} \quad (Equation 5.2)$$

Where:

ER_{Bovaer}	=	Enteric emissions reductions from lactating dairy cattle during the reporting period (MT CO ₂ e)
$BE_{Enteric}$	=	Baseline emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)
$PE_{Enteric}$	=	Total project enteric CH ₄ emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)

5.1 Bovaer Baseline Emissions

For Bovaer, baseline emissions are calculated according to Equation 5.3 in 2019 Refinement to the 2006 IPCC Guidelines for National GHG Inventories. Project emissions are calculated by determining the enteric methane emissions during the reporting period, adjusting those emissions in accordance with the Dose + NDF equation in Kebreab et al (2023), and adding emissions from the manufacturing and transport of Bovaer.

Baseline GHG emissions from enteric fermentation follow the Intergovernmental Panel on Climate Change (IPCC, 2019) Tier 2 methodology. Emissions for the baseline scenario are calculated according to the following:

$$BE_{Enteric} = CH_4 \times \frac{GWP}{1000} \quad (Equation 5.3)$$

Where:

$BE_{Enteric}$	=	Baseline emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)
CH_4	=	Baseline methane emissions for lactating dairy cattle during the reporting period (kg CH ₄)
GWP	=	Global warming potential of methane over a 100-year time frame; 27 is used per IPCC, 2021 (kg CO ₂ e per kg CH ₄)
1000	=	kg CO ₂ e per MT CO ₂ e

Baseline methane emissions for lactating dairy cattle during the reporting period are calculated according to the following:

$$CH_4 = \frac{(DMI \times GE) \times \left(\frac{Y_m}{100}\right) \times C \times t}{55.65} \quad (\text{Equation 5.4})$$

Where:

CH_4	=	Baseline enteric methane emissions for lactating dairy cattle during the reporting period (kg CH_4)
DMI	=	Average daily dry matter intake for lactating dairy cattle during the reporting period (kg DM per head per day)
GE	=	Gross energy concentration of lactating dairy cattle diet (MJ per kg dry matter)
Y_m	=	Methane conversion factor for lactating dairy cattle during the reporting period (percentage of dietary gross energy lost as CH_4 ; obtained from Table 6-1 adapted from Table 10.12 of IPCC, 2019)
C	=	Average number of total lactating dairy cattle during the reporting period (head)
t	=	Total number of days in the reporting period (days)
55.65	=	Energy content of methane (MJ per kg CH_4)

The appropriate Y_m is selected from **Table 5.1** Methane Conversion Factors (Y_m) for lactating dairy cattle from Table 10.12 of IPCC, 2019 based on diet nutrient composition. Nutrient composition is calculated using values from NASEM (2021) feed library or custom values based on the weighted averages of feed ingredients delivered, all documented in UpLook.

Table 5.1 Methane Conversion Factors (Y_m) for lactating dairy cattle from Table 10.12 of IPCC, 2019

Diet digestible energy (%)	Diet neutral detergent fiber (%)	Y_m
DE $\geq$ 70	NDF $\leq$ 35	5.7
DE $\geq$ 70	NDF $\geq$ 35	6.0
DE 63-70	NDF $>$ 37	6.3
DE $\leq$ 62	NDF $>$ 38	6.5

Please note Table 10.12 from IPCC, 2019 does not allow for appropriate Y_m selection guidance if DE $<$ 70% and NDF $\leq$ 37%. Therefore, if DE% and NDF% constraints do not allow for appropriate Y_m selection guidance with Table 6-1, a value of 5.85 is used.

A value of 5.85 was chosen because it is an average of the two tiers encompassing NDF $\leq$ 37%. This is accurate because this selection follows NDF constraints ($\leq$ 37% NDF) more than DE constraints, which agrees with other scientific equations and opinions that NDF is a major factor driving methane production (NASEM, 2021). The approach is also conservative because by following NDF constraints more closely than DE, a lower Y_m is used which results in smaller estimates of baseline methane and thus smaller estimates of carbon reductions due to the implementation of Bovaer.

5.2 Bovaer Project Emissions

Emissions in the project scenario are estimated as the sum of emissions during the reporting period from enteric fermentation and the production and transport of Bovaer according to the following equation:

$$PE_{Enteric} = CH_4 \times \left[1 - \frac{|AF_H|}{100} \right] \times \left[\frac{GWP}{1000} \right] + GHG_{MT} \quad (\text{Equation 5.5})$$

Where:

$PE_{Enteric}$	=	Total project enteric CH ₄ emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)
CH_4	=	Baseline methane emissions for lactating dairy cattle during the reporting period (kg CH ₄)
AF_H	=	Herd enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer; (Equation 5.6)
GWP	=	Global warming potential of methane over a 100-year time frame (kg CO ₂ e per kg CH ₄ ; IPCC, 2021)
1000	=	kg CO ₂ e per MT CO ₂ e
GHG_{MT}	=	Emissions associated with Bovaer manufacturing and transport during the reporting period (MT CO ₂ e)

5.2.1 Enteric Methane Adjustment Factor

The total lactating dairy cow herd enteric methane adjustment factor is determined by:

$$AF_H = AF \times PBCD \quad (\text{Equation 5.6})$$

Where:

AF_H	=	Herd enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer)
AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer (Equation 5.8)
$PBCD$	=	Percent Bovaer-fed cow-days of total cow-days available (%) (Equation 5.7)

Percent Bovaer-fed cow-days is calculated according to the following equation:

$$PBCD = \frac{\sum_i BC_i \times Bt_i}{C \times t} \quad (\text{Equation 5.7})$$

Where:

$PBCD$	=	Percent Bovaer-fed cow-days of total cow-days available (%)
BC_i	=	Average number of lactating dairy cattle in group i fed Bovaer during the reporting period (head)
Bt_i	=	Number of days in the reporting period Bovaer was fed to group i (days)

C	=	Average number of total lactating dairy cattle fed during the reporting period (head). If multiple groups of cattle included in one project, this is the sum of the average number of each group.
t	=	Total number of days in the reporting period (days)
i	=	Different groups of Bovaer-fed lactating dairy cattle where days and animal numbers differ between groups

The enteric methane adjustment factor due to Bovaer implementation is calculated according to the following:

$$AF = -32.8 - 0.285 \times (DOSE - 70.5) + 0.633 \times (NDF - 32.9) \quad (\text{Equation 5.8})$$

Where:

AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer)
-32.8	=	Overall mean estimate (intercept; % methane change) in Dose + NDF model in Table 2 of Kebreab et al. 2023
0.285	=	3-NOP dose factor in Dose + NDF model in Table 2 of Kebreab et al. 2023
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
70.5	=	3-NOP dose constant (mean dose; mg/kg) from Table 1 of Kebreab et al., 2023
0.633	=	NDF content factor from Dose + NDF model in Table 2 of Kebreab et al., 2023
NDF	=	Average diet neutral detergent fiber during the reporting period (% diet DM)
-32.9	=	NDF content constant (mean content; % DM) from Table 1 of Kebreab et al., 2023

5.2.2 GHG Emissions from Bovaer Manufacturing and Transport

Emissions from the Bovaer feed ingredients are estimated by including all GHG sources from manufacturing and transport according to the following:

$$GHG_{MT} = GHG_M + GHG_T \quad (\text{Equation 5.9})$$

Where:

GHG_{MT}	=	Emissions associated with Bovaer manufacturing and transport during the reporting period (MT CO ₂ e)
GHG_M	=	Emissions associated with Bovaer manufacturing during the reporting period (MT CO ₂ e)
GHG_T	=	Emissions associated with Bovaer transport during the reporting period (MT CO ₂ e)

The total amount of product used during the reporting period is calculated according to the following:

$$Bovaer10 = \frac{DMI \times DOSE}{0.1 \times 1000000} \times PBCD \times C \times t \quad (\text{Equation 5.10})$$

Where:

<i>Bovaer10</i>	=	Amount of fed product (10% 3-NOP) used during the reporting period (kg)
<i>DMI</i>	=	Average daily dry matter intake during the reporting period (kg DM per head per day)
<i>DOSE</i>	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
<i>PBCD</i>	=	Percent Bovaer-fed cow-days of total cow-days available (%)
<i>C</i>	=	Average number of lactating dairy cattle fed Bovaer during the reporting period (head)
<i>t</i>	=	Number of days in the reporting period (days)
0.1	=	Active ingredient content of Bovaer10 (kg 3-NOP per kg Bovaer10)
1000000	=	mg 3-NOP per kg 3-NOP

Manufacturing-based GHG emissions associated with Bovaer production are supplied by the manufacturer based on ISO standards 14040.44:2006 and ISO 14067:2018 and calculated according to the following:

$$GHG_M = \frac{Bovaer10 \times PCF}{1000} \quad (\text{Equation 5.11})$$

Where:

<i>GHG_M</i>	=	Manufacturing-based GHG emissions due to Bovaer use during the reporting period (MT CO ₂ e)
<i>Bovaer10</i>	=	Amount of Bovaer10 used during the reporting period according to (Equation 5.10)(kg)
<i>PCF</i>	=	Product Carbon Footprint of Bovaer10 production (4.84 kg CO ₂ e per kg Bovaer10)
1000	=	kg CO ₂ e per MT CO ₂ e

Emissions from the transport of the feed ingredient to the project site are calculated as:

$$GHG_T = \frac{Bovaer10 \times (EFT_R + EFT_O)}{1000} \quad (\text{Equation 5.12})$$

Where:

<i>GHG_T</i>	=	Emissions associated with Bovaer transport during the reporting period (MT CO ₂ e)
<i>Bovaer10</i>	=	Amount of Bovaer10 used during the reporting period according to (Equation 5.10)(kg)
<i>EFT_R</i>	=	Emissions factor from road transportation of Bovaer10 during the reporting period (0.6750 kg CO ₂ e per kg Bovaer10)
<i>EFT_O</i>	=	Emissions factor from ocean transportation of Bovaer10 during the reporting period (0.0812 kg CO ₂ e per kg Bovaer10)
1000	=	kg CO ₂ e per MT CO ₂ e

Road transportation of Bovaer is estimated using Environmental Protection Agency 2024 GHG Emissions Factors Hub data (Table 8; Scope 3 Category 4: Upstream Transportation and Distribution and Category 9: Downstream Transportation and Distribution) including CO₂, CH₄, and N₂O from medium- and heavy-duty truck emissions. Ocean transport is estimated using Department for Environment, Food and Rural Affairs, Annex 7 Freight Transport Conversion Tables and includes refrigerated cargo with the highest GHG emissions in order to be conservative. Distances of travel are considered the same for all farms implementing Bovaer and are thus conservatively estimated as the distance for product to be delivered to Tulare, CA. This location was chosen because it contains the largest population of cows the furthest distance from the Elanco warehouse in Clinton, IN.

5.3 Rumensin Project Emissions

$$ER_{Rumensin} = \frac{\sum_T ((GHG_B^T - GHG_P^T) \times PRCD_T)}{1000} \quad (\text{Equation 5.13})$$

Where:

$ER_{Rumensin}$	=	Net GHG emissions from the project after implementation of the practices (kg CO ₂ e)
T	=	Cattle type as specified in Table 4.3
GHG_B^T	=	Total GHG emissions for cattle type T from the baseline scenario (kg CO ₂ e)
GHG_P^T	=	Total GHG emissions for cattle type T from the project scenario (kg CO ₂ e)
$PRCD_T$	=	Percent Rumensin-fed Cattle-days for cattle type T
1000	=	kg CO ₂ e per MT CO ₂ e

Percent Rumensin-fed cattle-days is calculated according to the following equation:

$$PRCD_T = \frac{\sum_i RC_i \times Rt_i}{C \times t} \quad (\text{Equation 5.14})$$

Where:

$PRCD_T$	=	Percent Rumensin-fed Cattle-days for cattle type T
RC_i	=	Average number of cattle in group i fed Rumensin during the reporting period (head)
Rt_i	=	Number of days in the reporting period Rumensin was fed to group i (days)
C	=	Average number of total cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)
i	=	Different groups of Rumensin-fed cattle where days and animal numbers differ between groups

Rumensin project GHG emissions are calculated according to the following equation:

$$GHG_P^T = GHG_{feed,P}^T + GHG_{ent,P}^T + GHG_{man,P}^T \quad (\text{Equation 5.15})$$

Where:

GHG_P^T	=	Total GHG emissions for cattle type T from the project scenario (kg CO ₂ e)
$GHG_{feed,P}^T$	=	GHG emissions from the cultivation of cattle feed for cattle type T during the reporting period (kg CO ₂ e)
$GHG_{ent,P}^T$	=	GHG emissions from enteric fermentation from cattle type T during the reporting period (kg CO ₂ e). Lactating cattle type may not have a value for this.
$GHG_{man,P}^T$	=	GHG emissions from manure management for cattle type T during the reporting period (kg CO ₂ e)

Under this program $GHG_{ent,P}^T$ may only include cattle types "Dry Cattle" and "Replacement Heifers".

5.3.1 Feed Cultivation

Feed emissions in the project scenario are estimated as the sum of the emissions during the reporting period from the emissions due to cultivation of feed and the production and transport of Rumensin according to the following equations:

$$GHG_{feed,P}^T = (\sum_{i=1}^n EF_{feed\ i} \times DMI_{feed\ i}^{TOTAL,P}) + GHG_{MT} \quad (\text{Equation 5.16})$$

$$DMI_{feed\ i}^{TOTAL,P} = \sum_T (DMI_P^T \times C^T \times \varphi_{feed\ i}^T \times t) \quad (\text{Equation 5.17})$$

Where:

$GHG_{feed,P}^T$	=	GHG emissions for cattle type T from the cultivation of cattle feed during the reporting period (kg CO ₂ e)
$EF_{feed\ i}$	=	Emissions factor (kg CO ₂ e per kg DM) for feed i
$DMI_{feed\ i}^{TOTAL,P}$	=	Total dry matter intake of feed i to cattle type T during the reporting period (kg)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day)
C^T	=	Number of cattle fed during the reporting period for each animal type (head)
$\varphi_{feed\ i}^T$	=	Fractional makeup of feed i for cattle type T
t	=	Number of days in the reporting period (days)
GHG_{MT}	=	Emissions associated with Rumensin manufacturing and transport during the reporting period (kg CO ₂ e)

Emission factors for feed production per feed type shall be documented. All emission factors will be sourced from life cycle assessment databases, Ecoinvent 3.12² and Global Feed Lifecycle Assessment Institute (GFLI) 2025³. Given the broader and more established and internationally recognized presence of Ecoinvent, it will act as the primary database from which feed emission factors are sourced. In instances where both databases have the same feed ingredient emission factor available, preference will be given to the database with the most recently updated crop production data, as reported by both databases in their platform documentation.

If a specific emission factor is not available in Ecoinvent, GFLI will be leveraged as a secondary database. In cases where neither database contains a necessary emission factor, best professional judgement will be used and documented, prioritizing sourcing emissions factors from peer-reviewed and up-to-date scientific literature. Additional procedures include using economic allocation for co-products relative to an intended food/fuel product, when appropriate. Feed emissions factors should be relevant to North America, estimated using economic allocation, and include GHG emissions associated with cultivation, harvest, processing, transport, and upstream sources. Feed emissions factors will not differ between feedstuffs grown on-farm and purchased feedstuffs. Example GHG emissions from the cultivation of common feedstuffs are calculated using data from GFLI and are expressed in the sample calculation.

5.3.2 GHG Emissions from Rumensin Manufacturing and Transport

Emissions from the Rumensin feed ingredient is estimated by including all GHG sources from manufacturing and transport according to the following equation:

$$GHG_{MT} = GHG_M + GHG_T \quad (\text{Equation 5.18})$$

Where:

GHG_{MT}	=	Emissions associated with Rumensin manufacturing and transport during the reporting period (kg CO ₂ e)
GHG_M	=	Emissions associated with Rumensin manufacturing during the reporting period (kg CO ₂ e)
GHG_T	=	Emissions associated with Rumensin transport during the reporting period (kg CO ₂ e)

The total amount of product used during the reporting period is calculated according to the following equation:

$$Rumensin90 = \sum_T DOSE_P^T \times \frac{(DMI_P^T \times C^T \times t)}{(2000 \times 1000 \times 0.20)} \times PRCD_P^T \quad (\text{Equation 5.19})$$

Where:

$Rumensin90$	=	Amount of fed product (90.7 g per pound monensin) used during the reporting period (kg) for all cattle types T
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² ecoinvent (version 3.12, 2025), ecoinvent Association, Zurich Switzerland.

³ GFLI (2025), Global Feed LCA Institute, globalfeedlca.org

T	=	Cattle type
$DOSE_P^T$	=	Dose of Rumensin fed to Cattle type T during the reporting period (grams per ton per diet DM)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day)
C^T	=	Average number of cattle fed during the reporting period for each animal type (head)
t	=	Number of days in the reporting period (days)
2000	=	Pounds (lb) per ton
1000	=	grams (g) per kilogram (kg)
0.20	=	Percent active ingredient in Rumensin90
$PRCD_P^T$	=	Percent Rumensin-fed cow-days of total cow-days available (%) for cattle type T

Manufacturing-based GHG emissions associated with Rumensin90 production are based on ISO standards 14040.44:2006 and ISO 14067:2018 and calculated according to the following equation:

$$GHG_M = Rumensin90 \times PCF_R \quad (Equation 5.20)$$

Where:

GHG_M	=	Manufacturing-based GHG emissions due to Rumensin use in lactating dairy cattle during the reporting period (kg CO ₂ e)
$Rumensin90$	=	Amount of Rumensin90 used during the reporting period according to (Equation 5.19) (kg)
PCF_R	=	Product Carbon Footprint of Rumensin90 (kg CO ₂ e per kg Rumensin90)

Emissions from the transport of the feed ingredient to the project site are calculated as:

$$GHG_T = Rumensin90 \times EFT_{R,R} \quad (Equation 5.21)$$

Where:

GHG_T	=	Emissions associated with Rumensin transport during the reporting period (kg CO ₂ e)
$Rumensin90$	=	Amount of Rumensin90 used in lactating dairy cattle during the reporting period according to (Equation 5.19) (kg)
$EFT_{R,R}$	=	Emissions from road transportation of Rumensin90 total of 2,000 miles during the reporting period (0.4134 kg CO ₂ e per kg Rumensin90)

Road transportation of Rumensin is estimated using Environmental Protection Agency 2025 GHG Emissions Factors Hub data (Table 8; Scope 3 Category 4: Upstream Transportation and Distribution and Category 9: Downstream Transportation and Distribution) including CO₂, CH₄, and N₂O from medium- and heavy-duty truck emissions. Distances of travel are considered the same for all farms implementing Rumensin and are thus conservatively estimated as the

distance for product to be delivered to Tulare, CA. This location was chosen because it contains the largest population of cows the furthest distance from the Elanco manufacturing facility in Clinton, Indiana, United States. All Rumensin product is manufactured at the Clinton, Indiana facility, meaning calculated ocean transport emission factors are not applied.

5.3.3 GHG Emissions from Enteric Fermentation

These equations are only to be used for dry cattle and replacement heifers. ***Lactating Dairy Cattle consuming Bovaer may not quantify reductions according to these equations.***

Project GHG emissions from enteric fermentation for dry cattle and heifers follow the Intergovernmental Panel on Climate Change (IPCC), 2019, Tier 2 methodology. Diet nutrient composition (i.e., DE, NDF, etc.) should be documented and primary NDF data is encouraged where available for individual ingredients. Nutrient composition is calculated using feed ingredient composition weighted averages and feed values from either primary ingredient-level analysis (for NDF) or the 2021 NASEM feed library.⁴

$$GHG_{ent,P}^T = \sum_T (CH_4^T_{ent,P} \times GWP_{CH_4}) \quad (\text{Equation 5.22})$$

Where:

$GHG_{ent,P}^T$	=	GHG emissions from enteric fermentation from cattle type T during the reporting period (kg CO ₂ e)
$CH_4^T_{ent,P}$	=	Methane emissions from enteric fermentation for cattle type T during the reporting period (kg CH ₄)
GWP_{CH_4}	=	27; Global warming potential of methane (tCO ₂ e per tCH ₄) ⁵

The methane emissions generated through enteric fermentation are calculated as:

$$CH_4^T_{ent,P} = EF_{CH_4}^T_{ent,P} \times C^T \times t \quad (\text{Equation 5.23})$$

Where:

$CH_4^T_{ent,P}$	=	Methane emissions from enteric fermentation for cattle type T during the reporting period (kg CH ₄)
$EF_{CH_4}^T_{ent,P}$	=	Emission factor for methane of cattle type T during the reporting period (kg CH ₄ per head per day)
C^T	=	Number of cattle fed during the reporting period for cattle type T type (head)
t	=	Number of days in the reporting period (days)

⁴ National Academies of Sciences, Engineering, and Medicine. 2021. Nutrient Requirements of Dairy Cattle: Eighth Revised Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25806>

⁵ IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2391 pp.

The emission factor is calculated according to Equation 5.24⁶:

$$EF_{CH_4\ ent,P}^T = \frac{DMI_P^T \times GE \times \left(\frac{Y_{m,P}^T}{100}\right)}{55.65} \quad (\text{Equation 5.24})$$

Where:

$EF_{CH_4\ ent,P}^T$	=	Emission factor for methane of cattle type T during the reporting period (kg CH ₄ per head per day)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day, fresh feed only)
$Y_{m,P}^T$	=	Methane conversion factor during the reporting period from Methane Conversion Factor of Animals Fed Rumensin in Table 5.2 (percentage of dietary gross energy lost as CH ₄)
55.65	=	Energy content of methane (MJ per kg CH ₄)
GE	=	Gross energy concentration of diet (MJ per kg dry matter)

Table 5.2 Methane Conversion Factor by Cattle Type

Cattle Type	Feed quality Digestibility (DE %) and Neutral Detergent Fiber (NDF, % DMI) ⁷	2019 IPCC Methane Conversion Factor (Y _{m,B})	Change to Y _m with Feeding Rumensin	Methane Conversion Factor of Animals Fed Rumensin (Y _{m,P})
Lactating Dairy Cows	DE ≥ 70 NDF ≤ 35	5.7	-3.6%	5.49
	DE ≥ 70 NDF ≥ 35	6.0	-3.6%	5.78
	DE 63-70 NDF > 37	6.3	-3.6%	6.07
	DE ≤ 62 NDF > 38	6.5	-3.6%	6.27
Dry Cows		6.3	-3.6%	6.07
Replacement Heifers		6.3	-5.2%	5.97

DE= Diet digestible energy (%)

NDF= Diet neutral detergent fiber (%)

The change to Y_m (dietary gross energy lost as CH₄) with feeding Rumensin is calculated by dividing the Rumensin effect on dietary gross energy lost as CH₄ (intercept) in Model I of

⁶ Equation 10.21 from Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use

⁷ Please note Table 10.12 from IPCC, 2019 does not allow for appropriate Y_m selection guidance if DE <70% and NDF ≤ 37%. Therefore, if DE% and NDF% constraints do not allow for appropriate Y_m selection guidance with Table 5.1 (e.g., a diet with 69% DE and 36% NDF), a value of 5.85 is used. A value of 5.85 was chosen because it is an average of the two tiers encompassing NDF ≤ 37%. This is accurate because this selection follows NDF constraints (≤37% NDF) more than DE constraints, which agrees with other scientific equations and opinions that NDF is a major factor driving methane production (NASEM, 2021). The approach is also conservative because by following NDF constraints more closely than DE, a lower Y_m is used which results in smaller estimates of baseline methane and thus smaller estimates of carbon reductions due to the implementation of Bovaer.

Table 4 in Appuhamy et al. (2013) for each respective cattle type by the average dietary gross energy lost as CH₄ for both dairy cows and beef steers (6.35%; Y_m) from Table 3 in the Appuhamy paper, and is calculated as:

$$\% \text{ change in } Y_m = \frac{YMI_T}{6.35} \times 100 \quad (\text{Equation 5.25})$$

Where:

YMI_T	=	Intercept for Rumensin effect on Y _m of cattle type T ⁸ (percentage of dietary gross energy lost as CH ₄); Model I, Table 4 of Appuhamy et al., 2013
6.35	=	Control group Y _m average (percentage of dietary gross energy lost as CH ₄); Table 3 of Appuhamy et al., 2013

5.3.4 GHG Emissions from Manure Management

For manure management equations, accurate climate zone selection is critical to proper selection of IPCC factors. Climate Zone selection must be completed via the Elanco Climate Zone Tool. The climate zone utilized in the UpLook Calculator is determined through the Intergovernmental Panel on Climate Change (IPCC) 2006 guidelines for National Greenhouse Gas inventories with revisions in 2019. The guidelines mapped out in the IPCC created the allocations of climate zones utilizing elevation, mean annual temperature (MAT), mean annual precipitation (MAP), mean annual precipitation to potential evapotranspiration ratio (MAP:PET), and frost occurrence. The climate zone map generated by the IPCC guidelines & data from 1985 to 2015 following Harris et al., 2014. The map can be found in Volume 4: Chapter 3 Consistent representation of Lands.

The UpLook Climate Zone map is built using the IPCC Climate Zone Map's vectors. (Data is found in Q4: - 30 which includes dataset, raster, and vectors). In order to use the Climate Zone tool for a given herd, producers must:

1. Open map [link](#).
2. In top right-hand corner, select the eye glass.
3. Type in city & state or zip code of the site of the project.
4. Click on the city within the map to initiate popup with the allocated climate zone.

The output is then entered by the producer into UpLook. Producers are supported through the selection of their manure management systems with the UpLook Manure Management System eTool.

Project GHG emissions from manure production and management are calculated according to the following equations:

⁸ "Dairy cows" intercept of -0.23 is used for dry and lactating cows, "beef steers" intercept of -0.33 is used for replacement heifers and beef animals; Beef steer data are extrapolated to replacement heifers because 1) there is no biological reason sex would alter Rumensin's mode of action, 2) the studies contributing to the data set encompass a range of high and low roughage diets, and 3) adjustments to Y_m were calculated based on a model adjusting for dry matter intake which can differ across cattle types and ages.

$$GHG_{man,P}^T = \sum_T \left((CH_4^T_{man,P} \times GWP_{CH_4}) + (N_2O^T_{man,P} \times GWP_{N_2O}) \right) \quad (\text{Equation 5.26})$$

Where:

$GHG_{man,P}^T$	=	GHG emissions from manure management for cattle type T during the reporting period (kg CO ₂ e)
$CH_4^T_{man,P}$	=	Methane emissions from manure production for cattle type T during the reporting period (kg CH ₄)
GWP_{CH_4}	=	27; Global warming potential of methane (tCO ₂ e per tCH ₄)
$N_2O^T_{man,P}$	=	Nitrous oxide emissions from manure production for cattle type T during the reporting period (kg N ₂ O)
GWP_{N_2O}	=	273; Global warming potential of nitrous oxide (kg CO ₂ e per kg N ₂ O) ⁹

5.3.4.1 Manure Methane Emissions

Project methane emissions for the management of manure are calculated using a Tier 2 approach consistent with both IPCC and Environmental Protection Agency (EPA) quantification methodologies. It also leverages the approach used by the Climate Action Reserve's U.S. Livestock Project Protocol. Diet nutrient composition (i.e., DE, NDF, etc.) should be documented and primary NDF data is encouraged where available for individual ingredients. Nutrient composition is calculated using feed ingredient composition weighted averages and feed values from either primary ingredient-level analysis (for NDF) or the 2021 NASEM feed library.¹⁰

$$CH_4^T_{man,P} = \sum_{S,T} (MS_{S,P}^T) \quad (\text{Equation 5.27})$$

Where:

$CH_4^T_{man,P}$	=	Methane emissions from manure production for cattle type T during the reporting period (kg CH ₄)
S	=	Manure treatment system
$MS_{S,P}^T$	=	Methane emissions from storage/treatment system S by cattle type T, aggregated for the reporting period (kg CH ₄)

$$MS_{S,P}^T = (C^T \times PS_{S,P}^T \times VS_P^T \times MCF_S \times B_0^T \times \rho_{CH_4} \times t) \quad (\text{Equation 5.28})$$

Where:

$MS_{S,P}^T$	=	Methane emissions from storage/treatment system S by cattle type T, aggregated for the reporting period (kg CH ₄)
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⁹ IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2391 pp.

¹⁰ National Academies of Sciences, Engineering, and Medicine. 2021. Nutrient Requirements of Dairy Cattle: Eighth Revised Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25806>

C^T	=	Number of cattle fed during the reporting period for cattle type T (head)
$PS_{S,P}^T$	=	Percent of manure sent to (managed in) manure storage/treatment system S from cattle type T in the reporting period (%)
VS_P^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period (kg per head per day)
MCF_S	=	Methane conversion factor for storage/treatment system S from Table 5.3 (%)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T in Table 5.4 (m3 CH ₄ per kg VS)
ρ_{CH_4}	=	0.67; Density of methane = 0.67 at 1 atm and 60°F (kg per m ³) ¹¹
t	=	Number of days in the reporting period (days)

$$VS_P^T = \left[(DMI_P^T \times GE) \times \left(1 - \frac{DE_P^T}{100} \right) + (UE_P^T \times [DMI_P^T \times GE]) \right] \times \left[\frac{1 - ASH_P^T}{GE} \right] \quad (\text{Equation 5.29})$$

Where:

VS_P^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period (kg per head per day)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day)
GE	=	Gross energy concentration of diet (MJ per kg dry matter)
DE_P^T	=	Average diet digestibility of the feed for cattle type T in the reporting period (% of gross energy) ¹²
UE_P^T	=	4%; Urinary energy as a percent of gross energy of cattle type T in the reporting period (%) ¹³
ASH_P^T	=	8%; Ash content of feed for cattle type T in the reporting period (%) ^{14,15}

¹¹ From Equation 10.23 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use

¹² Diet digestibility calculated using values from National Academies of Sciences, Engineering, and Medicine. 2021. Nutrient Requirements of Dairy Cattle: Eighth Revised Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25806>

¹³ From Equation 10.24 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use

¹⁴ From Equation 10.24 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use

¹⁵ Dämmgen U. et al. (2011). Reassessment of the calculation procedure for the volatile solids excretion rates of cattle and pigs in the Austrian, Danish and German agricultural emission inventories. Landbauforschung Volkenrode 61:115-126.

Table 5.3 Methane conversion factor (MCF) for manure management systems¹⁶

System		MCFs by Climate Zone									
		Cool				Temperate		Warm			
		Cool Temperate Moist	Cool Temperate Dry	Boreal Moist	Boreal Dry	Warm Temperate Moist	Warm Temperate Dry	Tropical Montane	Tropical Wet	Tropical Moist	Tropical Dry
Uncovered Anaerobic Lagoon		60%	67%	50%	49%	73%	76%	76%	80%	80%	80%
Liquid/Slurry, And Pit Storage Below Animal Confinements	1 Month	6%	8%	4%	4%	13%	15%	25%	38%	36%	42%
	3 Month	12%	16%	8%	8%	24%	28%	43%	61%	57%	62%
	4 Month	15%	19%	9%	9%	29%	32%	50%	67%	64%	68%
	6 Month	21%	26%	14%	14%	37%	41%	59%	76%	73%	74%
	12 Month	31%	42%	21%	20%	55%	64%	73%	80%	80%	80%
Cattle And Swine Deep Bedding (Cont.)	> 1 Month	21%	26%	14%	14%	37%	41%	59%	76%	73%	74%
Cattle And Swine Deep Bedding	< 1 Month	2.75%				6.50%		18%			
Solid Storage		2.00%				4.00%		5.00%			
Solid Storage – Covered/Compacted		2.00%				4.00%		5.00%			
Solid Storage – Bulking Agent Addition		0.50%				1.00%		1.50%			
Solid Storage – Additives		1.00%				2.00%		2.50%			
Dry Lot		1.00%				1.50%		2.00%			
Daily Spread		0.10%				0.50%		1.00%			
Composting – In-Vessel		0.50%									
Composting – Static Pile (Forced Aeration)		1.00%				2.00%		2.50%			
Composting – Intensive Windrow		0.50%				1.00%		1.5%			
Composting – Passive Windrow (Infrequent Turning)		1.00%				2.00%		2.50%			
Pasture/Range/Paddock		0.47%									
Poultry Manure With And Without Litter		1.50%									
Aerobic Treatment		0.00%									
Burned For Fuel		10.00%									

¹⁶ Adapted from Table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

System	MCFs by Climate Zone									
	Cool				Temperate		Warm			
	Cool Temperate Moist	Cool Temperate Dry	Boreal Moist	Boreal Dry	Warm Temperate Moist	Warm Temperate Dry	Tropical Montane	Tropical Wet	Tropical Moist	Tropical Dry
Anaerobic Digester, Low Leakage, High Quality Gastight Storage, Best Complete Industrial Technology	1.00%									
Anaerobic Digester, Low Leakage, High Quality Industrial Technology, Low Quality Gastight Storage Technology	1.41%									
Anaerobic Digester, Low Leakage, High Quality Industrial Technology, Open Storage	3.55%				4.38%		4.59%			
Anaerobic Digester, High Leakage, Low Quality Technology, High Quality Gastight Storage Technology	9.59%									
Anaerobic Digester, High Leakage, Low Quality Technology, Low Quality Gastight Storage Technology	10.85%									
Anaerobic Digester, High Leakage, Low Quality Technology, Open Storage	12.14%				12.97%		13.17%			

Table 5.4 Maximum methane producing capacity of manure for cattle¹⁷

Cattle Types (T)	B ₀ T ¹ (m ³ CH ₄ kg ⁻¹ VS)
Dry And Lactating Dairy Cows	0.24
Replacement Dairy Heifers	0.19

5.3.4.2 Manure Nitrous Oxide Emissions

Manure nitrous oxide emissions are calculated according to IPCC and the guidance of International Dairy Federation, 2020, which includes a minimum requirement of "direct and indirect N₂O emissions from excreta in cattle housing, storage and treatment."¹⁸ Project emissions of N₂O are calculated as follows

¹⁷ From Table 10.16 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use

¹⁸ IDF. 2022. The IDF global Carbon Footprint standard for the dairy sector. In: Bulletin of the IDF No. 520/2022. International Dairy Federation (ed.), Brussels.

$$N_2O_{man,P}^T = \sum_T (N_2O_{d,P}^T + N_2O_{i,l,P}^T + N_2O_{i,v,P}^T) \quad (\text{Equation 5.30})$$

Where:

$N_2O_{man,P}^T$	=	Nitrous oxide emissions from manure production for cattle type T during the reporting period (kg N ₂ O)
$N_2O_{d,P}^T$	=	Direct N ₂ O emissions from manure management for cattle type T during the reporting period (kg N ₂ O)
$N_2O_{i,l,P}^T$	=	Indirect N ₂ O emissions from leaching of manure for cattle type T during the reporting period (kg N ₂ O)
$N_2O_{i,v,P}^T$	=	Indirect N ₂ O emissions from the volatilization of NH ₃ and NO _x for cattle type T during the reporting period (kg N ₂ O)

5.3.4.2.1 Direct N₂O Emissions from Manure Management

The calculation of direct N₂O emissions entails multiplying the total amount of nitrogen excretion in each type of manure management system by an emission factor for that type of system and aggregating the values. The emissions are then calculated using the following equations:

$$N_2O_{d,P}^T = (\sum_S N_{S,P}^T) \times EF_{d,S} \times \frac{44}{28} \quad (\text{Equation 5.31})$$

$$N_{S,P}^T = (N_{ex,P}^T \times PS_{S,P}^T) \quad (\text{Equation 5.32})$$

Where:

$N_2O_{d,P}^T$	=	Direct N ₂ O emissions from manure management for cattle type T during the reporting period (kg N ₂ O)
$N_{S,P}^T$	=	Total nitrogen from cattle type T managed by storage system S during the reporting period (kg N)
$EF_{d,S}$	=	Emissions factor for direct N ₂ O emissions from the manure management system S in Table 5.5 (kg N ₂ O-N per kg N excreted)
$44/28$	=	Conversion of N ₂ O-N emissions to N ₂ O emissions
$N_{ex,P}^T$	=	N excretion by cattle type T, aggregated for the reporting period (kg N)
$PS_{S,P}^T$	=	Percent of manure sent to (managed in) manure storage/treatment system S from cattle type T in the reporting period (%)

Table 5.5 Default emission factors for direct N₂O Emission from manure management¹⁹

System	Definition	EFd,S [kg N ₂ O-N (kg N excreted) ⁻¹]
Pasture/Range/ Paddock	The manure from pasture and range grazing animals is allowed to lie as is and is not managed.	Direct and indirect N ₂ O emissions associated with the manure deposited on agricultural soils and pasture, range, and paddock systems are not included in this protocol.
Daily Spread	Manure is routinely removed from a confinement facility and is applied to cropland or pasture within 24 hours of excretion. N ₂ O emissions during storage and treatment are assumed to be zero. N ₂ O emissions from land application are covered under the Agricultural Soils category.	0
Solid Storage	The storage of manure, typically for a period of several months, in unconfined piles or stacks. Manure is able to be stacked due to the presence of a sufficient amount of bedding material or loss of moisture by evaporation.	0.010
Solid Storage – Covered/ Compacted	Similar to solid storage, but the manure pile is a) covered with a plastic sheet to reduce the surface of manure exposed to air and/or b) compacted to increase the density and reduce the free air space within the material.	0.01
Solid Storage – Bulking Agent Addition	Specific materials (bulking agents) are mixed with the manure to provide structural support. This allows the natural aeration of the pile, thus enhancing decomposition. (e.g. sawdust, straw, coffee husks, maize stover)	0.005
Solid Storage – Additives	The addition of specific substances to the pile in order to reduce gaseous emissions. Addition of certain compounds such as attapulgite, dicyandiamide or mature compost have shown to reduce N ₂ O emissions; while phosphogypsum reduce CH ₄ emission	0.005
Dry Lot	A paved or unpaved open confinement area without any significant vegetative cover where accumulating manure may be removed periodically. Dry lots are most typically found in dry climates but also are used in humid climates.	0.02
Liquid/Slurry	Manure is stored as excreted or with some minimal addition of water to facilitate handling and is stored in either tanks or earthen ponds.	With natural crust cover
		Without natural crust cover
		Cover

¹⁹ Adapted from Table 10.21 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

System	Definition	EF _{d,S} [kg N ₂ O-N (kg N excreted) ⁻¹]
Uncovered Anaerobic Lagoon	Anaerobic lagoons are designed and operated to combine waste stabilization and storage. Lagoon supernatant is usually used to remove manure from the associated confinement facilities to the lagoon. Anaerobic lagoons are designed with varying lengths of storage (up to a year or greater), depending on the climate region, the volatile solids loading rate, and other operational factors. The water from the lagoon may be recycled as flush water or used to irrigate and fertilize fields.	0
Pit Storage Below Animal Confinements	Collection and storage of manure usually with little or no added water typically below a slatted floor in an enclosed animal confinement facility.	0.002
Anaerobic Digester	Anaerobic digesters are designed and operated for waste stabilization by the microbial reduction of complex organic compounds to CH ₄ and CO ₂ , which is captured and flared or used as a fuel.	0.0006
Deep Bedding	As manure accumulates, bedding is continually added to absorb moisture over a production cycle and possibly for as long as 6 to 12 months. This manure management system also is known as a bedded pack manure management system and may be combined with a dry lot or pasture.	No mixing 0.01
		Active mixing 0.07
Composting – In-Vessel	Composting, typically in an enclosed channel, with forced aeration and continuous mixing.	0.006
Composting – Static Pile (Forced Aeration)	Composting in piles with forced aeration but no mixing.	0.010
Composting – Intensive Windrow (Frequent Turning)	Composting in windrows with regular turning for mixing and aeration.	0.005
Composting – Passive Windrow (Infrequent Turning)	Composting in windrows with infrequent turning for mixing and aeration.	0.005
Aerobic Treatment	The biological oxidation of manure collected as a liquid with either forced or natural aeration. Natural aeration is limited to aerobic and facultative ponds and wetland systems and is due primarily to photosynthesis. Hence, these systems typically become anoxic during periods without sunlight.	Natural aeration systems 0.01
		Forced aeration systems 0.005

Nitrogen intake is used to determine N excretion and is calculated for all cattle types according to the following equation²⁰:

²⁰ Equation 10.32 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

$$N_{i,P}^T = DMI_P^T \times \frac{CP^T/100}{6.25} \quad (\text{Equation 5.33})$$

Where:

$N_{i,P}^T$	=	Nitrogen intake for cattle type T during the reporting period (kg per head per day)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day)
CP^T	=	Crude protein content in the diet for cattle type T (%) ²¹
6.25	=	Conversion from kg of dietary protein to kg of dietary N (kg feed protein per kg N)

Rate of N excretion for lactating cows is calculated according to the following equation²²:

$$N_{ex,P}^T = (N_{i,P}^T - N_{R,P}^T) \times C^T \times t \quad (\text{Equation 5.34})$$

Where:

$N_{ex,P}^T$	=	N excretion by cattle type T, aggregated for the reporting period (kg N)
$N_{i,P}^T$	=	Nitrogen intake for cattle type T during the reporting period (kg per head per day)
$N_{R,P}^T$	=	N retained by lactating animals. Calculated according to (Equation 5.35) (kg per head per day)
C^T	=	Number of cattle fed during the reporting period for cattle type T(head)
t	=	Number of days in the reporting period (days)

Nitrogen retained by lactating animals is calculated according to the following equation²³:

$$N_{R,P}^T = \left[\frac{M \times \left(\frac{M_p}{100} \right)}{6.38} \right] + \left[\frac{WG \times \left[268 - \left(\frac{7.03 \times NE_G}{WG} \right) \right]}{1000 \times 6.25} \right] \quad (\text{Equation 5.35})$$

Where:

$N_{R,P}^T$	=	N retained by lactating animals (kg per head per day)
M	=	Milk production (kg per head per day)
M_p	=	Percent of protein in the milk (%)
6.38	=	Conversion from milk protein to milk N (kg protein per kg N)

²¹ From NASEM, 2021. Nutrient Requirements of Dairy Cattle: Eighth Revised Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25806>

²² Equation 10.31A in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

²³ Equation 10.33 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

WG	=	Weight gain; 0 for mature lactating cattle ²⁴
NE_G	=	Net energy for growth (MJ/d); not applicable for lactating mature lactating cattle because weight gain = 0

Rate of N excretion for all other cattle types (dry cows and replacement heifers) is calculated according to the following equation²⁵:

$$N_{ex,P}^T = N_{i,P}^T \times (1 - N_{R_frac}^T) \times C^T \times t \quad (\text{Equation 5.36})$$

Where:

$N_{ex,P}^T$	=	N excretion by cattle type T, aggregated for the reporting period (kg N)
$N_{i,P}^T$	=	Nitrogen intake for cattle type T during the reporting period (kg per head per day)
$N_{R_frac}^T$	=	N retention fraction in non-lactating cattle types from Table 5.6 Data for estimating N retention fractions for various cattle types (kg N retained per head per day per kg N intake per head per day)
C^T	=	Number of cattle fed during the reporting period for cattle type T (head)
t	=	Number of days in the reporting period (days)

Table 5.6 Data for estimating N retention fractions for various cattle types²⁶

Animal Type	Fraction N Retained (kg N retained/animal/day) (kg N intake/animal/day) ⁻¹
Dry Cows	0.07
Replacement Heifers	0.14

5.3.4.2.2 Indirect N₂O Emissions from Leaching of Manure

Nitrogen is lost through runoff and leaching into soils from the solid storage of manure at outdoor areas and in feedlots. The amount of nitrous oxide emitted through leaching shall be calculated using the following equation:

$$N_2O_{i,l,P}^T = \sum_S (N_{S,P}^T \times \text{Frac}_{leach_{MS}}^T) \times EF_5 \times \frac{44}{28} \quad (\text{Equation 5.37})$$

Where:

²⁴ Table 10A.1 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

²⁵ Equation 10.31 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

²⁶ Table 10A.2 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

$N_2O_{i,l,P}^T$	=	Indirect N ₂ O emissions from leaching of manure for cattle type T during the reporting period (kg N ₂ O)
$N_{S,P}^T$	=	Total nitrogen from cattle type T managed by storage system S during the reporting period (kg N)
$Frac_{leach_{MS_S}}^T$	=	Fraction of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in manure management system S from Table 5.7
EF_5	=	Emission factor for N ₂ O emissions from nitrogen leaching and runoff from Table 5.8 (kg N ₂ O -N per kg N leached and runoff)
$44/28$	=	Conversion of N ₂ O-N emissions to N ₂ O emissions

5.3.4.2.3 Indirect N₂O Emissions from Volatilization of NH₃ and NO_x

Nitrogen in the volatilized form of ammonia may be deposited at sites downwind from manure handling areas and contribute to indirect N₂O emissions. The amount of nitrous oxide emitted through volatilization in the forms of NH₃ and NO_x is calculated using the following equation:

$$N_2O_{i,v,P}^T = \sum_S (N_{S,P}^T \times Frac_{Gas_{MS_S}}^T) \times EF_4 \times \frac{44}{28} \quad (\text{Equation 5.38})$$

Where:

$N_2O_{i,v,P}^T$	=	Indirect N ₂ O emissions from the volatilization of NH ₃ and NO _x for cattle type T during the reporting period (kg N ₂ O)
$N_{S,P}^T$	=	Total nitrogen from cattle type T managed by storage system S during the reporting period (kg N)
$Frac_{Gas_{MS_S}}^T$	=	Fraction of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x in the manure management system S from Table 5.7
EF_4	=	Emission factor for N ₂ O emissions from nitrogen volatilization from Table 5.8 (kg N ₂ O - N per kg NH ₃ -N + NO _x -N volatilized)
$44/28$	=	Conversion of N ₂ O-N emissions to N ₂ O emissions

Table 5.7 Default values for N loss fractions due to volatilization of NH₃ and NO_x and leaching of N from manure management²⁷

System	FracLeach_MS	FracGas_MS
Uncovered Anaerobic Lagoon	0	0.35
Liquid/Slurry - With Natural Crust Cover	0	0.30
Liquid/Slurry - Without Natural Crust Cover	0	0.48
Liquid/Slurry - With Cover	0	0.10
Pit Storage Below Animal Confinements	0	0.28
Daily Spread	0	0.07
Solid Storage	0.02	0.30
Solid Storage - Covered/Compacted	0	0.14
Solid Storage - Bulking Agent Addition	0.02	0.38
Solid Storage - Additives	0.02	0.11

²⁷ From Table 10.22 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Dry Lot	0.035	0.30
Anaerobic Digester	0	0.05-0.50
Cattle And Swine Deep Bedding	0.035	0.25
Composting - In-Vessel	0	0.45
Composting - Static Pile	0.06	0.50
Composting - Intensive Windrow	0.06	0.50
Composting - Passive Windrow	0.04	0.45
Aerobic Treatment - Natural Aeration Systems	0	No data
Aerobic Treatment - Forced Aeration Systems	0	0.85

Table 5.8 Default emission volatilization and leaching factors used for indirect soil N₂O emissions²⁸

Climate	EF ₄ , kg N ₂ O-N (kg NH ₃ -N + NO _x - N volatilized) ⁻¹	EF ₅ , kg N ₂ O-N/kg N leached and runoff
Cool Temperate Moist	0.014	0.011
Cool Temperate Dry	0.005	0.011
Boreal Moist	0.014	0.011
Boreal Dry	0.005	0.011
Warm Temperate Moist	0.014	0.011
Warm Temperate Dry	0.005	0.011
Tropical Montane	0.014	0.011
Tropical Wet	0.014	0.011
Tropical Moist	0.014	0.011
Tropical Dry	0.005	0.011

5.4 Rumensin Baseline GHG Emissions

Baseline emissions for Rumensin are a quantification of the GHG emissions from sources within the GHG Assessment Boundary that would have occurred in the absence of the project. Project emissions are the actual GHG emissions that occur at sources within the GHG Assessment Boundary during the reporting period. Project emissions must be subtracted from baseline emissions to quantify the project's total net GHG emission reductions.

Baseline emissions for Rumensin are calculated using (Equation 5.39) through (Equation 5.56) and the adjustment factors in Table 5.9.

Equation	Equation #	Description
$GHG_B^T = GHG_{feed,B}^T + GHG_{ent,B}^T + GHG_{man,B}^T$	(Equation 5.39)	Total GHG emissions for cattle type T in the baseline scenario (kg CO ₂ e)

²⁸ From Table 11.3 in Chapter 11 of the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

$GHG_{feed,B}^T = \sum_{i=1}^n EF_{feed\ i} \times \left(\frac{DMI_{feed\ i}^{TOTALP}}{AF_{DMI}^T} \right)$	(Equation 5.40)	GHG emissions from the cultivation of cattle feed for cattle type T in the baseline scenario (kg CO ₂ e)
$GHG_{ent,B}^T = \sum_T (CH_4^T_{ent,B} \times GWP_{CH_4})$	(Equation 5.41)	GHG emissions from enteric fermentation from cattle type T in the baseline scenario (kg CO ₂ e)
$CH_4^T_{ent,B} = EF_{CH_4\ ent,B}^T \times C^T \times t$	(Equation 5.42)	Methane emissions from enteric fermentation for cattle type T in the baseline scenario (kg CH ₄)
$EF_{CH_4\ ent,B}^T = \frac{\left(\frac{DMI_P^T}{AF_{DMI}^T} \right) \times GE \times \left(\frac{Y_{m,B}^T}{100} \right)}{55.65}$	(Equation 5.43)	Emission factor for methane of cattle type T in the baseline scenario (kg CH ₄ per head per day)
$GHG_{man,B}^T = \sum_T ((CH_4^T_{man,B} \times GWP_{CH_4}) + (N_2O^T_{man,B} \times GWP_{N_2O}))$	(Equation 5.44)	GHG emissions from manure management for cattle type T in the baseline scenario (kg CO ₂ e)
$CH_4^T_{man,B} = \sum_{S,T} (MS_{S,B}^T)$	(Equation 5.45)	Methane emissions from manure production for cattle type T in the baseline scenario (kg CH ₄)
$MS_{S,B}^T = (C^T \times PS_{S,p}^T \times VS_B^T \times MCF_S \times B_0^T \times \rho_{CH_4} \times t)$	(Equation 5.46)	Methane emissions from storage/treatment system S by cattle type T in the baseline scenario (kg CH ₄)
$VS_B^T = \left[\left(\frac{DMI_P^T}{AF_{DMI}^T} \right) \times GE \right] \times \left(1 - \frac{DE_P^T}{100} \right) + \left(UE_P^T \times \left[\left(\frac{DMI_P^T}{AF_{DMI}^T} \right) \times GE \right] \right) \times \left[\frac{(1-ASH_P^T)}{GE} \right]$	(Equation 5.47)	Volatile solids produced by cattle type T on a dry matter basis in the baseline scenario (kg per head per day)
$N_2O^T_{man,B} = \sum_T (N_2O_{d,B}^T + N_2O_{i,l,B}^T + N_2O_{i,v,B}^T)$	(Equation 5.48)	Nitrous oxide emissions from manure production for cattle type T in the baseline scenario (kg N ₂ O)
$N_2O_{d,B}^T = (\sum_S N_{S,B}^T) \times EF_{d,S} \times \frac{44}{28}$	(Equation 5.49)	Direct N ₂ O emissions from manure management for cattle type T in the baseline scenario (kg N ₂ O)
$N_2O_{i,l,B}^T = \sum_S (N_{S,B}^T \times \text{Frac}_{leach_MS_S}^T) \times EF_5 \times \frac{44}{28}$	(Equation 5.50)	Indirect N ₂ O emissions from leaching of manure for cattle type T in the baseline scenario (kg N ₂ O)
$N_2O_{i,v,B}^T = \sum_S (N_{S,B}^T \times \text{Frac}_{Gas_MS_S}^T) \times EF_4 \times \frac{44}{28}$	(Equation 5.51)	Indirect N ₂ O emissions from the volatilization of NH ₃ and NO _x for cattle type T in the baseline scenario (kg N ₂ O)
$N_{S,B}^T = (N_{ex,B}^T \times PS_{S,p}^T)$	(Equation 5.52)	Total nitrogen from cattle type T managed by storage system S in the baseline scenario (kg N)
$N_{ex,B}^T = (N_{i,B}^T - N_{r,B}^T) \times C^T \times t$	(Equation 5.53)	N excretion for lactating cattle type aggregated in the baseline scenario (kg N)

$N_{ex,B}^T = N_{i,B}^T \times (1 - N_{R,frac}^T) \times C^T \times t$	(Equation 5.54)	N excretion by non-lactating cattle types aggregated in the baseline scenario (kg N)
$N_{R,B}^T = \left[\frac{\frac{M}{AF_M^T} \times \left(\frac{M_p}{100} \right)}{6.38} \right] + \left[\frac{WG \times \left[\frac{268 - \left(\frac{7.03 \times NE_G}{WG} \right)}{1000} \right]}{6.25} \right]$	(Equation 5.55)	N retained by lactating animal in the baseline scenario (kg per head per day)
$N_{i,B}^T = \left(\frac{DMI_P^T}{AF_{DMI}^T} \right) \times \left[\frac{\left(\frac{CP^T}{100} \right)}{6.25} \right]$	(Equation 5.56)	Nitrogen intake for cattle type T in the baseline scenario (kg per head per day)

Where:

GHG_B^T	=	Total GHG emissions for cattle type T from the baseline scenario (kg CO ₂ e)
$GHG_{feed,B}^T$	=	GHG emissions for cattle type T from the cultivation of cattle feed in the baseline scenario (kg CO ₂ e)
$GHG_{ent,B}^T$	=	GHG emissions from enteric fermentation from cattle type T in the baseline scenario (kg CO ₂ e)
$GHG_{man,B}^T$	=	GHG emissions from manure management for cattle type T in the baseline scenario (kg CO ₂ e)
$EF_{feed\ i}$	=	Emissions factor (kg CO ₂ e per kg DM) for feed i
$DMI_{feed\ i}^{TOTAL,P}$	=	Total dry matter intake of feed i to cattle type T during the reporting period (kg)
AF_{DMI}^T	=	Dry matter intake adjustment factor for cattle type T from Table 5.9
$CH_4_{ent,B}^T$	=	Methane emissions from enteric fermentation for cattle type T in the baseline scenario (kg CH ₄)
GWP_{CH_4}	=	27; Global warming potential of methane (tCO ₂ e per tCH ₄)
$EF_{CH_4\ ent,B}^T$	=	Emission factor for methane of cattle type T in the baseline scenario (kg CH ₄ per head per day)
C^T	=	Number of cattle fed during the reporting period for each animal type (head)
t	=	Number of days in the reporting period (days)
DMI_P^T	=	Average daily dry matter intake for cattle type T during the reporting period (kg DM per head per day)
$Y_{m,B}^T$	=	Methane conversion factor in the baseline scenario in Table 5.2 (percentage of dietary gross energy lost as CH ₄)
55.65	=	Energy content of methane (MJ per kg CH ₄)
GE	=	Gross energy concentration of diet (MJ per kg dry matter)
$CH_4_{man,B}^T$	=	Methane emissions from manure production for cattle type T in the baseline scenario (kg CH ₄)
$N_2O_{man,B}^T$	=	Nitrous oxide emissions from manure production for cattle type T in the baseline scenario (kg N ₂ O)
GWP_{N_2O}	=	273; Global warming potential of nitrous oxide (kg CO ₂ e kg N ₂ O ⁻¹)

$MS_{S,B}^T$	=	Methane emissions from storage/treatment system S by cattle type T in the baseline scenario (kg CH ₄)
$PS_{S,P}^T$	=	Percent of manure sent to (managed in) manure storage/treatment system S from cattle type T in the reporting period (%)
VS_B^T	=	Volatile solids produced by cattle type T on a dry matter basis in the baseline scenario (kg per head per day)
MCF_S	=	Methane conversion factor for storage/treatment system S from Table 5.3 (%)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T in Table 5.4 (m ³ CH ₄ per kg VS)
ρ_{CH_4}	=	0.67; Density of methane = 0.67 at 1 atm and 60°F (kg per m ³)
DE_P^T	=	Average diet digestibility of the feed for cattle type T in the reporting period (% of gross energy)
UE_P^T	=	4%; Urinary energy as a percent of gross energy of cattle type T in the reporting period (%)
ASH_P^T	=	8%; Ash content of feed for cattle type T in the reporting period (%)
$N_2O_{d,B}^T$	=	Direct N ₂ O emissions from manure management for cattle type T in the baseline scenario (kg N ₂ O)
$N_2O_{i,l,B}^T$	=	Indirect N ₂ O emissions from leaching of manure for cattle type T during in the baseline scenario (kg N ₂ O)
$N_2O_{i,v,B}^T$	=	Indirect N ₂ O emissions from the volatilization of NH ₃ and NO _x for cattle type T in the baseline scenario (kg N ₂ O)
$N_{S,B}^T$	=	Total nitrogen from cattle type T managed by storage system S in the baseline scenario period (kg N)
$EF_{d,S}$	=	Emissions factor for direct N ₂ O emissions from the manure management system S in Table 5.5 (kg N ₂ O-N per kg N excreted)
$44/28$	=	Conversion of N ₂ O-N emissions to N ₂ O emissions
$Frac_{leach_MS_S}^T$	=	Fraction of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in manure management system S from Table 5.7 .
EF_5	=	Emission factor for N ₂ O emissions from nitrogen leaching and runoff from Table 5.8 (kg N ₂ O-N per kg N leached and runoff)
$Frac_{Gas_MS_S}^T$	=	Fraction of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x in the manure management system S from Table 5.7
EF_4	=	Emission factor for N ₂ O emissions from nitrogen volatilization from Table 5.8 (kg N ₂ O-N per kg NH ₃ -N + NO _x -N volatilized)
$N_{ex,B}^T$	=	N excretion for cattle type T aggregated in the baseline scenario (kg N)
$N_{i,B}^T$	=	Nitrogen intake for cattle type T in the baseline scenario (kg per head per day)
$N_{R,B}^T$	=	N retained by lactating dairy cow type through milk in the baseline scenario (kg per head per day)
$N_{R_frac}^T$	=	N retention fraction in non-lactating cattle types from Table 5.6 (kg N retained per head per day per kg N intake per head per day)
M	=	Milk production during the reporting period (kg per head per day)
M_p	=	Percent of protein in the milk (%)
6.38	=	Conversion from milk protein to milk N (kg protein per kg N)
AF_M^T	=	Milk yield adjustment factor for lactating cows from Table 5.9

WG	=	Weight gain; 0 for mature lactating cattle
NE_G	=	Net energy for growth (MJ/d); not applicable for lactating mature lactating cattle because weight gain = 0
CP^T	=	Crude protein content in the diet for cattle type T (%)
6.25	=	Conversion from kg of dietary protein to kg of dietary N (kg feed protein per kg N)

Table 5.9 Baseline adjustment factors

Adjustment Factor	Cattle Type	Value	Reference
AF_{DMI}^T	Dry Dairy Cows	0.977	Duffield et al., 2008
AF_{DMI}^T	Lactating Dairy Cows	0.977	Duffield et al., 2008
AF_{DMI}^T	Replacement Dairy Heifers	0.957	Gadberry et al., 2022
AF_M^T	Lactating Dairy Cows	1.005	Duffield et al., 2008

The baseline adjustment factor for dry matter intake is calculated according to the following equation:

$$AF_{DMI}^T = 1 - \left| \frac{DM_change_T}{100} \right| \quad (Equation 5.57)$$

Where:

AF_{DMI}^T	=	Dry matter intake adjustment factor for cattle type T
DM_change_T	=	Change in dry matter intake in cattle type T due to Rumensin (%)

Because significant heterogeneity was observed with milk components, the reported yield and component percentages reported in Table 2 of Duffield et al., 2008 were used to calculate solids-corrected milk (SCM) differences between control and Rumensin treatments according to Vasquez et al. 2021, resulting in a 0.5% increase in SCM with Rumensin. This 0.5% increase was utilized for Rumensin's effect on milk yield. The adjustment factor for milk yield is calculated according to the following equation:

$$AF_M^T = 1 + \left| \frac{SCM_R - SCM_C}{SCM_C} \right| \quad (Equation 5.58)$$

Where:

AF_M^T	=	Milk yield adjustment factor for lactating cows
SCM_R	=	Solids-corrected milk in Rumensin cows (kg/d)
SCM_C	=	Solids-corrected milk in control cows (kg/d)

Solids corrected milk in control cows is calculated according to the following equation:

$$SCM_C = MY_C \times [(12.24 \times fat\%_C \times 0.01) + (7.1 \times protein\%_C \times 0.01) + (6.35 \times lactose\%_C \times 0.01) - 0.0345] \quad (Equation 5.59)$$

Values for control milk yield (kg/d), fat %, protein %, and lactose % are calculated by dividing the raw weighted mean difference (Rumensin – control) by the percent change from Duffield et al. (2008) according to the following equations:

$$MY_C = \frac{MY_dif}{MY_change} \quad (Equation 5.60)$$

$$fat\%_C = \frac{fat\%_dif}{fat\%_change} \quad (Equation 5.61)$$

$$protein\%_C = \frac{protein\%_dif}{protein\%_change} \quad (Equation 5.62)$$

$$lactose\%_C = \frac{lactose\%_dif}{lactose\%_change} \quad (Equation 5.63)$$

Where:

SCM_C	= Solids-corrected milk in control cows (kg/d)
MY_C	= Control milk yield (kg/d)
$fat\%_C$	= Control milk fat (%)
$protein\%_C$	= Control milk protein (%)
$lactose\%_C$	= Control milk lactose (%)
MY_dif	= Milk yield, raw weighted mean difference; 0.7 kg/d
MY_change	= Milk yield, percent change; 2.3%
$fat\%_dif$	= Milk fat percent, raw weighted mean difference; -0.12
$fat\%_change$	= Milk fat percent, percent change; -3.1%
$protein\%_dif$	= Milk protein percent, raw weighted mean difference; -0.03
$protein\%_change$	= Milk protein percent, percent change; -0.9%
$lactose\%_dif$	= Milk lactose percent, raw weighted mean difference; -0.012
$lactose\%_change$	= Milk lactose percent, percent change; -0.25%

Solids corrected milk in Rumensin cows is calculated according to the following equation:

$$SCM_C = MY_C \times [(12.24 \times fat\%_R \times 0.0.1) + (7.1 \times protein\%_R \times 0.01) + (6.35 \times lactose\%_R)] \quad (Equation 5.64)$$

Values for Rumensin milk yield (kg/d), fat %, protein %, and lactose % are calculated by adding raw weighted mean differences from the calculated control values according to the following equations:

$$MY_R = MY_C + MY_dif \quad (Equation 5.65)$$

$$fat\%_R = fat\%_C + fat\%_dif \quad (Equation 5.66)$$

$$protein\%_R = protein\%_C + protein\%_dif \quad (Equation 5.67)$$

$$lactose\%_R = lactose\%_C + lactose\%_dif \quad (Equation 5.68)$$

Where:

MY_R	=	Rumensin milk yield (kg/d)
MY_C	=	Control milk yield (kg/d)
MY_dif	=	Milk yield, raw weighted mean difference; 0.7 kg/d
$fat\%_C$	=	Control milk fat (%)
$protein\%_C$	=	Control milk protein (%)
$lactose\%_C$	=	Control milk lactose (%)
$fat\%_R$	=	Rumensin milk fat (%)
$protein\%_R$	=	Rumensin milk protein (%)
$lactose\%_R$	=	Rumensin milk lactose (%)
$fat\%_dif$	=	Milk fat percent, raw weighted mean difference; -0.12
$protein\%_dif$	=	Milk protein percent, raw weighted mean difference; -0.03
$lactose\%_dif$	=	Milk lactose percent, raw weighted mean difference; -0.012

5.5 Fat and Protein Corrected Milk

The good impacted by this intervention is fat and protein corrected milk (FPCM). This is quantified as follows:

$$FPCM = Milk\ Shipped \times [0.1226 \times fat(\%) \times 0.0776 \times protein(\%) + 0.2534] \quad (Equation 5.69)$$

Where:

<i>FPCM</i>	=	Fat and Protein Corrected Milk from lactating dairy cattle during the reporting period (kg)
<i>Milk Shipped</i>	=	Quantity of milk shipped (kg) to processor during the reporting period
0.1226	=	Fat coefficient
<i>fat</i> (%)	=	% fat shipped to processor during the reporting period
0.0776	=	Protein coefficient
<i>protein</i> (%)	=	% true protein shipped to processor during the reporting period. If true protein is not available, it may be estimated using crude protein shipped minus 0.19.
0.2534	=	Intercept

The choice of FPCM as the good is driven by market needs of potential offtake partners. FPCM has been identified as the most used metric for emissions factors comparisons used by engaged offtake partners. The equation above comes from the 2022 International Dairy Federation Guidelines, Equation 1.

5.6 Leakage & Permanence

During the development of this protocol, the need to account for market-shifting leakage associated with changes in productivity, feed, and manure was evaluated. Bovaer does not significantly impact productivity or dry matter intake (feed requirements), as evidenced by the 5 meta-analyses conducted. Bovaer is metabolized by its own mode of action (Duin et al., 2016) and is not found in manure. Studies between control and 3-NOP-fed animals have shown no differences in fecal characteristics (volatile solids, carbon, nitrogen or total solids) in forage- or grain-based diets (Nkemka et al., 2019), no differences in manure GHG and NH₃ emissions or manure application to crop yield (Owens et al., 2020, 2021), and no impact on anaerobic digestion (Nkemka et al., 2019). Therefore, because Bovaer does not affect any GHG emission sources outside the identified GHG boundaries, leakage is not relevant to this project and no deductions will be applied to the quantified reductions.

Leakage related to Rumensin was also considered. The current project carries no risk of more than 5% reduction in crop production for any specific feed component due to the data supporting quantification of project feed cultivation GHG reductions. Because the maximum reduction in feed production within this protocol is less than 5%, leakage is not relevant, and no deductions will be applied to credits generated according to this protocol.

Emissions reductions associated with feeding Bovaer and Rumensin are permanent and cannot be reversed.

5.7 Uncertainty

A quantitative uncertainty assessment was conducted according to SustainCert guidelines. In accordance with GHG Protocol scope 3 Quantitative Inventory Uncertainty guidance, a pedigree matrix approach was used to quantify uncertainty where a parameter uncertainty was unknown.

A Monte Carlo simulation was conducted to propagate parameter uncertainty to determine the overall uncertainty on the adjustment factor (AF) due to implementation of Bovaer. This includes the SEM reported as ± 1.6 for the intercept, ± 0.074 for Dose, ± 0.252 for NDF

parameters from Kebreab et al. (2023). The Monte Carlo simulation yielded an expected methane percent change of -31.60%, accompanied by a 90% confidence interval ranging from -34.82% (upper bound) to -28.41% (lower bound) and an uncertainty of 10.1% (calculated as (upper bound – lower bound) / (2 x expected)). This equates to a mean CO₂e change (project minus baseline) of -1.46 MT CO₂e with a 90% confidence interval ranging from -1.31 (lower bound) to -1.61 (upper bound) MT CO₂e annually and an uncertainty of 10.1%.

The major risk that could substantially affect the project's GHG emission reductions is in the estimation of feed intake. Traditionally a large amount of uncertainty in dairy operation greenhouse gas calculations originates from estimations of feed intake and digestibility, which vary widely between and within cattle and geographies. This uncertainty is mitigated through the use of each project's individual feed ingredient data to quantify dry matter intake, more accurately estimate gross and digestible energy, and to calculate dose. The availability of these data facilitates a precise measure of gross energy intake from which enteric emissions reductions are calculated.

The remaining uncertainty associated with the GHG reductions quantified in this protocol arises from the calculation of the adjustment factor as a result of Bovaer implementation. These adjustments are taken directly from peer-reviewed meta-analyses. Bovaer has reliable and substantiated data regarding its mode of action and effect as described throughout this document.

Uncertainty pertaining to determination of NDF is mitigated by allowing inclusion of feed-analysis derived NDF values for specific ingredients. This ensures that where NASEM values are used, their use is consistent; and where primary data is available, it is collected in a specific and credible manner.

Because uncertainty associated with feed intake is mitigated through rigorous monitoring and incorporating project-specific feed data in a tier 2 approach to calculate GHG emissions and uncertainty associated with adjustment factors for both Bovaer and Rumensin's effects are mitigated using a meta-analysis, no deductions for uncertainty will be taken.

5.8 Deviations from Protocol Methodologies

Deviations from the methodologies in section 5 of this protocol are not allowed.

6 Monitoring

This program is 100% monitoring. All producers participating in the program will go through verification of their baseline data and verification of monitoring periods. A monitoring plan has been developed for all monitoring and reporting activities associated with the project, standardized across all participating farms.

Verifiers will use the monitoring plan and report to confirm that the requirements of this program have been met. This monitoring plan provides the processes, requirements, and sources of information necessary to assess the GHG reductions created by the practices included in this protocol.

This includes:

1. The procedures for collecting data on intervention activities related to implementation.
2. The data points collected to verify emission reduction, project, and baseline calculations.
3. The QC/QA processes to ensure the accuracy and consistency of the data collected.

The monitoring reports described in the monitoring plan include the following elements:

1. General description of the project, including the location of the cattle operations
2. List of the practices implemented
3. Description of the process and frequency of data collection and the archiving procedures
4. Recordkeeping plan
5. Role of any individuals performing activities related to the practices implemented
6. Quality assurance/quality control (QA/QC) procedures to ensure the accurate collection and entry of data in quantification systems
7. Monitoring reports must include the monitoring time period.
8. Monitoring reports must include the list of parameters measured and monitored.
9. Monitoring reports must include the types of data and information reported, including units of measurement.
10. Monitoring reports must include the origin of the data.
11. The monitoring report must include an attestation as to regulatory compliance.
12. The monitoring report should be submitted no less frequently than annually and no more frequently than 30 days.
13. The monitoring period can be as short as 30 days. The maximum monitoring period is 12 months.
14. The monitoring report must be submitted and shared with Athian, as the program administrator.

A monitoring period represents a calendar month. Where necessary, multiple months may be combined at the discretion of the verifier. When the monitoring period is over (i.e. the month has fully passed), data from operating records are input into the Athian quantification tool to assess the impact of the intervention for that monitoring period. Supporting documents are collected concurrently. Once all required inputs and supporting documentation are collected, a third-party verifier receives the information to assess the validity of the reported inputs as well as verify the quantification themselves. Participating producers are given the opportunity to produce further documentation of any values the verifier determines to be non-compliant before a final decision on the results of the monitoring period is rendered, either verified or not.

This monitoring plan provides the requirements and sources of information necessary to assess the GHG reductions created using this protocol. This monitoring plan describes the procedures for collecting data on intervention activities related to the implementation of

protocol practices. The data collected will support emission reduction and baseline calculations. This monitoring plan also outlines the QC/QA processes to ensure the accuracy and consistency of the data collected that will be used to verify emissions reduction outcomes.

6.1 Data Quality Assurance

Site specific data will be collected via UpLook, a digital tool which enables quantification, QA/QC, and monitoring. Reductions from implementation of Bovaer and Rumensin are calculated using a counterfactual approach based on each farm's individual data gathered through UpLook and integrated into Athian's platform. The two main factors contributing to enteric methane emissions are the number of cows and the total feed they consume. The next important factor is the type of diet the cows are consuming (i.e., diet composition and characteristics of feed ingredients). For Rumensin, primary data on feed ingredients and volumes fed as well as the active manure systems for each site will also be collected in Uplook.

Data sources utilized in UpLook™ Dairy as it relates to Bovaer intervention include both manually entered primary data entered by the user inputs correlated to the dairy operation along with digital data transferred from herd records (e.g., average number of lactating dairy cows in the period, percent of cows receiving Bovaer, Bovaer dose, and total feed intake by ingredient) and secondary table values that are internal to UpLook (e.g., NASEM feed ingredient characteristics, Ym values, PCF of Bovaer).

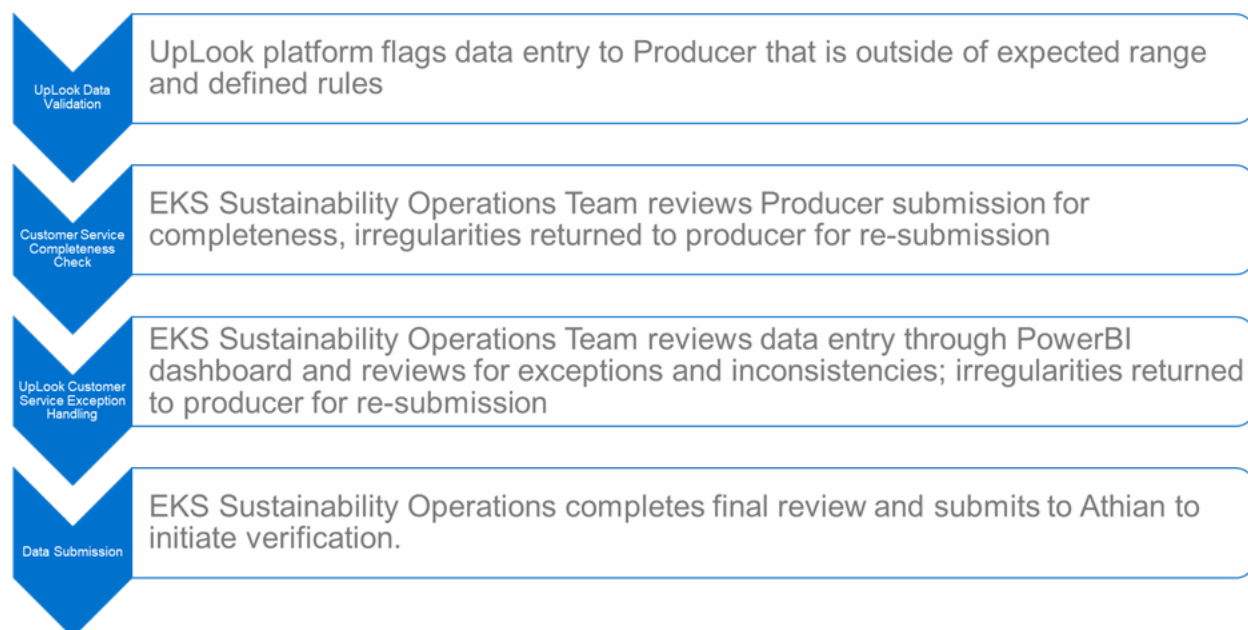
Elanco is responsible for the data quality in the UpLook tool. Elanco is responsible for any changes to the monitoring data and establishing and implementing quality control procedures.

Athian directly ingests data without alteration from the UpLook API endpoint on a daily basis. This process is confirmed via Athian's software Quality Assurance process involving both automated and manual testing to confirm expected functionality. Additionally, data is verified via the monitoring and verification processes detailed in Section 8.

6.1.1 UpLook Data Quality Management

All data entered in UpLook are subject to the Elanco Quality Assurance and Quality Control process prior to submission to Athian, **Figure 6.1** outlines the key steps. This process reduces the amount of bias and uncertainties and provides a high level of accuracy when quantifying reductions. Diet factors (i.e. NDF) that may impact the reduction are accounted for when quantifying the reduction. This provides reliable results maintained over a range of variable situations (diet types, manure management systems, etc) and reduces biases and uncertainties as far as are practical. This protocol allows for robust entry of primary data, and use of conservative secondary data when primary data is not available.

Figure 6.1 Elanco Quality Assurance Process



6.1.2 Athian Data Quality Management Plan

The Athian Data Quality Management Plan aims to ensure that a producer's data is accurate, reliable, and fit for its intended purpose to assess the impact of the practices included in this protocol on CO₂e emissions associated with the management of manure and feed cultivation. The goals and objectives of a can be categorized into several key areas, each targeting different aspects of data quality management. These include accuracy, timeliness, comparability, and creditability.

Accuracy:

- **Data Collection Methods:** Data will be provided by the farm directly based on various on farm systems.
- **Consistency Checks:** Input forms will check for data type and range preventing grossly invalid data from being entered.
- **Method Validation:** Based on type, input may be limited to certain ranges or values. Additionally, producers must attest to and confirm accuracy.

Timeliness:

- **Data Collection Frequency:** As defined by the protocol
- **Data Reporting Schedule:** Specific schedules are defined by the protocol monitoring plan.
- **Response Procedures for Data Variations:** Significant data issues should be prevented at entry. Additionally identified issues can be corrected by Athian staff as needed.

Comparability:

- Standardized Methods Used: Form input is used to collect quantitative data from on farm systems.
- Benchmarking: Per the Athian Data Retention Policy, all GHG related data is kept for a minimum of 10 years. Data, in aggregated and anonymized form, can be used for benchmarking if/when applicable.

Creditability:

- Documentation of Data Processes: All data processes are ultimately governed by the Athian Data Protection, Data Retention, and Software Development Lifecycle (SDLC) policies. These policies are maintained as controlled documents in the Athian compliance system (Drata) and are reviewed and updated at least annually.
- Transparency Measures: Data transparency is critical to credibility and integral to the data collection process. Data input directly in the platform from producers requires an attestation from the producer as to the accuracy before being submitted for verification. Data collected via an integration to a 3rd party data collection software also requires the producer to attest to the accuracy. In addition, producers have visibility as to the data provided to 3rd party verifiers and can see the status of the verification of each element of data submitted. All verification reports include each data element collected and reviewed as part of the verification process for complete transparency in the reporting of the emissions result.

The Athian platform has a comprehensive set of automated processes that confirm the integrity, correctness, and completeness of data. These include checking the data upon ingestion from any 3rd party data source, inclusive of data delivered via API or manually entered, for completeness and accuracy. These checks include verification of appropriate formatting, field-level requirements that ensure the presence of all required data, and identification of any data variance from the previously verified data. If errors are identified, notifications are generated and delivered to engineering, product management, and service management for resolution. Those parties then determine the source and scope of the issue(s), engage any necessary participating party, resolve and document the identified issues.

In addition to the data validation checks identified above, Athian has implemented a service-driven approach for applying logic consistently, significantly reducing the potential for error in the process. The programmatic logic used reduces or replaces much of the process that is prone to human error. The Athian platform hosts the mechanisms for documenting any data discrepancy as well as their respective severity and solution. The platform retains a complete transaction history for all data ingested, inclusive of date/time stamp and the individual user or software supplying the information. This ensures that Athian will have a complete history/picture of all data used when rendering a decision or result.

All data used to meet GHG carbon accounting standards for impact units must be retained for a minimum of ten years. This includes producer business contact information, location information, monitoring period information, and all verification information. All of the aforementioned processes and procedures adhere to industry best practices, including SOC 2 review. The quantification tool for this program is thoroughly tested against known results of data sets any time updates to the quantification methodology or tool are made. The tests follow the same methods as used for the Simulated Quantification Model and are checked against the quantification methodology for accuracy by the Athian development team.

6.2 Product Quality Assurance

6.2.1 Bovaer Product Quality

Bovaer is formulated into the lactating cow total mixed ration by a professional dairy nutritionist who is legally required to follow any directions listed on the Bovaer label. Dairy nutritionists often use ration balancing software (e.g., Nutritional Dynamic Systems (NDS) or Agricultural Modeling and Training Systems (AMTS)) as a tool to formulate and evaluate total mixed ration and premix recipes for the correct amounts of macronutrients (carbohydrates, protein, fat), micronutrients (vitamins and minerals), and feed ingredients. Nutritionists normally use ration software to ensure the correct amount of Bovaer is in the formulation. Within these ration balancing programs there is a library of feed ingredients, each with chemical composition documented. As an aid, Elanco has created a Bovaer feed ingredient specifications file for common ration balancing software to provide an easy-to-use ingredient profile for their recipe. Once these recipes are formulated by the dairy nutritionist, the recipe for the mix goes to a mill for manufacturing.

For quality control measures, mills storing and handling Bovaer must follow Code of Federal Regulations 21 CFR Part 507 (entitled “Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventative Controls for Food for Animals”; <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-E/part-507>), which includes following the guidelines of the Food Safety Modernization Act (FSMA). After the mix is delivered from the mill to the farm, the dairy producer is responsible for following the total mixed ration recipe formulated by the professional dairy nutritionist when loading and feeding their cattle. Please note that every farm’s situation is unique and there may be instances where Bovaer may be mixed into a premix on farm or delivered into the ration via a micro ingredient machine. Ultimately, the selection and implementation of the Bovaer dose incorporated into the ration is the responsibility of both the producer and the professional dairy nutritionist.

6.2.2 Rumensin Product Quality

Rumensin manufacturing processes are highly controlled and repeatable. The target potency for Rumensin 90 (Type A feed) is 90.7 g/lb of monensin. Approximately 98% of Rumensin batches are within $\pm 3\%$ of the designated target^[1], well within the $\pm 15\%$ allowed by the FDA. Additionally, strict dust control ensures that $>90\%$ of Rumensin particles are within the target range for adequate mixing (180 to 850 microns). Furthermore, Elanco includes microtracers in all Rumensin 90 sold in the United States. Microtracers provide a quick method for confirmation of inclusion or absence of Rumensin in supplements or feeds. Their use to validate feed mixing is included in the Standard of the American Society of Agricultural Engineers.

Like Bovaer, Rumensin is formulated into the dairy diet by a professional dairy nutritionist who is legally required to feed Rumensin according to the FDA approved label, often using ration balancing software as noted above. Ration balancing software includes a library of feed ingredients (like Rumensin) with documented chemical compositions to ensure accurate formulation. Dairy nutritionists and producers typically supplement Rumensin through a premix (Type B feed). Once these recipes are formulated by the dairy nutritionist, the Type B recipe goes to a mill for manufacturing.

For quality control measures, mills storing and handling Rumensin90 must follow Code of Federal Regulations 21 CFR Part 507 (entitled “Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventative Controls for Food for Animals”; <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-E/part-507>), which includes following the guidelines of the Food Safety Modernization Act (FSMA). After the Type B feed is delivered from the mill to the farm, the dairy producer is responsible for following the diet formulated by the professional dairy nutritionist when mixing the final diet (Type C feed) and feeding their cattle. Ultimately, the selection and implementation of the

Rumensin dose incorporated into the ration is the responsibility of both the producer and the professional dairy nutritionist.

Table 6.1 Monitoring parameters

Data/ Parameter	Description	Data Unit	Values Applied	Data Source	Measurement Frequency	Measurement methods and procedures, including QA/QC	Roles and Responsible	Data Management
<i>Milk Shipped</i>	Quantity of milk shipped to Processor during the reporting period	kg	Total volume of milk shipped to Processor based on milk checks or processor reporting associated with cattle in project during monitoring period. Even if only partial herds are included in the project, total volume of milk shipped to processor engaged in project should be included.	Operating records	Every reporting period	Data entered into UpLook & optionally supporting documentation attached (milk check or report from Processor).	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>Fat Shipped</i>	% fat of milk shipped to Processor during the reporting period	%	Total % of milk shipped to Processor based on milk checks or processor reporting associated with cattle in project during monitoring period, ranging from 3-7%.	Operating records	Every reporting period	Data entered into UpLook & optionally supporting documentation attached (milk check or report from Processor).	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>Protein Shipped</i>	% protein of milk shipped to Processor during the reporting period	%	Total % protein of milk shipped to Processor based on milk checks or processor reporting associated with cattle in project during monitoring period, ranging from 2.5 - 7%	Operating records	Every reporting period	Data entered into UpLook & optionally supporting documentation attached (milk check or report from Processor).	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

<i>M</i>	Milk production per animal	kg per head per day	Average daily production per day per lactating cow, ranging from 40 – 110 lbs/hd/day	Operating records	Every reporting period	Data entered into UpLook & optionally supporting documentation attached (milk check or report from Processor).	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>C</i> or <i>C^T</i>	Average number of cattle by type during the monitoring period	head	Farm inventory is calculated as the average number of lactating cows per month included in the project during the monitoring period	Operating records	Every reporting period	Input data for herd population numbers can be obtained from dairy farm records or herd management software.	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>T</i>	Type of cattle included in the project	N/A	N/A	Operating records	Every reporting period	N/A	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>t</i>	Number of days in the reporting period (calendar month)	Days	Number of days in month	UpLook	Every reporting period	N/A	Elanco: Responsible for UpLook that establishes the number of days for the month being analyzed	Data and optionally supporting documentation are stored in UpLook.

<i>PBCD</i>	Percent Bovaer-fed cow-days	Cow-days	Calculated as total Bovaer-fed cow-days divided by total cow-days available in reporting period (month). Total Bovaer-fed cow days accounts for the number of cows fed Bovaer and the number of days they were fed Bovaer during the reporting period.	Operating records & UpLook Tool 3: Percent Cow Days Calculator	Every reporting period	Calculation completed by producer, attached in UpLook with supporting evidence. QA/QC checks entry in UpLook against Tool 3 output and supporting documentation.	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>PRCD</i>	Percent Rumensin-fed cow-days	Cow-days	Calculated as total Rumensin -fed cow-days divided by total cow-days available in reporting period (month). Total Rumensin -fed cow days accounts for the number of cows fed Rumensin and the number of days they were fed Rumensin during the reporting period.	Operating records & UpLook Tool 3: Percent Cow Days Calculator	Every reporting period	Calculation completed by producer, attached in UpLook with supporting evidence. QA/QC checks entry in UpLook against Tool 3 output and supporting documentation.	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

<i>DMI or DMI_P^T</i>	Average dry mass of feed consumed by an animal(s) on a given day(s) during the reporting period, fresh feed only.	kg DM per head per day by cattle type; entered as average lb/hd/day, total lbs/mo, or total tons/month, UpLook converts to kg/hd/d	A weighted average of fresh feed ingredients delivered to each cattle type during the reporting period is used to estimate dry matter intake of each cattle type. Heifers: 5 to 25 lbs/head/day Dry: 15 to 45 lbs/head/day Lactating: 35 to 70 lbs/head/day	Operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>DE</i>	Diet digestible energy as a % of diet gross energy by cattle type	% of gross energy	Diet digestible energy composition is documented in UpLook using feed ingredient composition weighted averages and digestible energy values from the NASEM (2021) feed library	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating DE value of diet / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>NDF</i>	Neutral detergent fiber; A measure of the fiber concentration of the lactating dairy cattle diet	% of diet dry matter	Diet neutral detergent composition is documented in UpLook using feed ingredient composition weighted averages and neutral detergent fiber values from the NASEM (2021) feed library.	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating NDF value of diet / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

GE	Gross energy concentration of the diet by cattle type	MJ per kg diet dry matter	Diet gross energy composition is documented in UpLook using feed ingredient composition weighted averages, feed chemical compositions from the NASEM (2021) feed library, and Equations 3-1 and 3-2 from NASEM (2021).	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating GE value of diet / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
CP^T	Crude protein content in the diet for cattle type T (%)	%	Crude Protein % is documented in UpLook using feed ingredient composition weighted averages, feed chemical compositions from the NASEM (2021)	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating CP value of diet / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
Y_m	Methane conversion factor; Percentage of gross energy converted to methane	N/A	The appropriate Y_m is selected from Table 5.1 based on diet digestible energy (DE) and neutral detergent fiber (NDF) of the diet	Secondary data from IPCC, 2019; adapted to Table 5.1	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for selecting correct Y_m based on calculated DE and NDF values	Data and optionally supporting documentation are stored in UpLook.
$Y_{m,p}^T$	Methane conversion factor during the reporting period; Percentage of gross energy converted to methane	N/A	The appropriate Y_m is selected from Table 5.2 based on diet digestible energy (DE) and neutral detergent fiber (NDF) of the diet	Secondary data from IPCC, 2019;	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for selecting correct Y_m based on calculated DE and NDF values	Data and optionally supporting documentation are stored in UpLook.

$Y_{m,B}^T$	Methane conversion factor during the baseline scenario; Percentage of gross energy converted to methane	N/A	The appropriate Ym is selected from Table 5.2 based on diet digestible energy (DE) and neutral detergent fiber (NDF) of the diet	Secondary data from IPCC, 2019;	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for selecting correct Ym based on calculated DE and NDF values	Data and optionally supporting documentation are stored in UpLook.
$EF_{feed\ i}$	Emission factor for feed i	kg CO2e per kg DM	Appropriate value based on the feed ingredient is selected from either database dependent on data availability specific to the feed ingredient	Secondary data from GFLI or Ecolnvent	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for contracting with and maintaining a mapped library of feed emission factors from the GFLI and/or Ecolnvent databases; developing custom emission factors in cases of custom ingredients	Data and optionally supporting documentation are stored in UpLook.
$\varphi_{feed\ i}^T$	Fractional make up for feed i for cattle type T	%	Appropriate value based on the feed ingredient is selected from either database dependent on data availability specific to the feed ingredient	Operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
$GWP\ or\ GWP_{CH4}\ or\ GWP_{N2O}$	Global warming potential over 100-year time horizon	kg CO2e per kg CH4; kg CO2e per kg N2O; kg CO2e per kg	27 for Methane; 273 for Nitrous Oxide	Secondary data from Sixth Assessment Report from IPCC, 2021 (AR6)	Annually	Values reviewed during QA/QC process	Elanco: Responsible for updating value in UpLook when deemed necessary according to IPCC standards	Data stored in UpLook.

<i>AF</i>	Enteric methane adjustment factor due to Bovaer implementation	%	Calculated value using Dose + NDF model in Table 2 from the Kebreab et al., 2023 meta-analysis of 3-NOP effects on enteric methane in lactating dairy cattle and incorporating dose (primary farm data) and NDF (calculated value OR primary farm data) for each reporting period	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating value / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
<i>AF_{DMI}^T</i>	Dry matter intake adjustment factor for cattle type T	%	Appropriate value selected from Table 5.9	Secondary data from literature	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for calculating value	Data stored in UpLook.
<i>AF_M^T</i>	Milk yield adjustment factor for cattle type T	%	Appropriate value selected from Table 5.9	Secondary data from literature	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for calculating value	Data stored in UpLook.
<i>Dose</i>	Dose of 3-nitrooxypropanol (3-NOP) delivered to lactating cows; Bovaer10 is 10% 3-NOP	mg 3-NOP per kg fresh diet dry matter	Any value allowed, but if value is not in scientific range based on Kebreab et al., 2023, (37-80 mg/kg DM), then AF is set equal to zero.	Operating records & UpLook Tool 4: Bovaer Dose Calculator	Every reporting period	Calculation completed by producer, attached in UpLook with supporting evidence. QA/QC checks entry in UpLook against Tool 4 output and supporting documentation.	Producer: inputting these values and ensuring they are correct / Elanco: Responsible for calculating value / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

<i>PCF</i>	Carbon footprint associated with manufacturing Bovaer10	kg CO2e per kg Bovaer10	4.84	Secondary data originating from manufacturer	Annually	Value follows documented PCF completed by DNV; values reviewed in QA/QC process	Elanco: responsible for reviewing/updating product carbon footprint as needed	Data stored in UpLook.
<i>EFT_R</i>	Carbon footprint associated with land transportation of Bovaer10	kg CO2e per metric-ton mile for Bovaer10	0.18668	Calculated data originating from secondary sources	Annually	Values reviewed in QA/QC process	Elanco: responsible for reviewing/updating product transportation carbon footprint as needed	Data stored in UpLook.
<i>EFT_O</i>	Carbon footprint associated with ocean transportation of Bovaer10	kg CO2e per metric-ton mile for Bovaer10	0.06994	Calculated data originating from secondary sources	Annually	Values reviewed in QA/QC process	Elanco: responsible for reviewing/updating product transportation carbon footprint as needed	Data stored in UpLook.
<i>PS_{S,P}^T</i>	Percent of manure sent to (managed in) manure storage/treatment system S from cattle type T in the reporting period (%)	%	% of manure flowing through each system reported	Operating records & UpLook Tool 4: Manure Management System Wizard	Every reporting period	Calculation completed by producer, attached in UpLook with supporting evidence. QA/QC checks entry in UpLook against Tool 4 output and supporting documentation.	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for calculating % value for each system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

MCF_S	Methane conversion factor for storage/treatment system S	N/A	Producer determines their Climate Zone using Tool 5: Climate Zone Selection Tool; based on Climate Zone and storage / treatment system S UpLook determines MCF based on Table 5.3	Secondary data from IPCC, 2019; see Table 5.3	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining MCF based on system and climate zone / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
B_O^T	Maximum methane-producing capacity of manure for cattle type T	m3 CH4 per kg VS	Selected value from Table 5.4	Secondary data from IPCC, 2019; see Table 5.4	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for correctly selecting value based on Cattle Type T	Data stored in UpLook.
DE_P^T	Diet digestible energy as a % of diet gross energy by cattle type	% of gross energy	Diet digestible energy composition is documented in UpLook using feed ingredient composition weighted averages and digestible energy values from the NASEM (2021) feed library	Calculated value using primary data from operating records	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for calculating DE value of diet / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
UE_P^T	Urinary energy as a % of gross energy for cattle type T in the reporting period	%	4%	Secondary data from IPCC, 2019	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for maintaining references for UE	Data stored in UpLook.

ASH_P^T	Ash content of feed for cattle type T in the reporting period	%	8%	Secondary data from IPCC, 2019	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Elanco: Responsible for maintaining references for ASH	Data stored in UpLook.
$EF_{d,S}$	Emissions factor for direct N ₂ O emissions from the manure management system S	kg N ₂ O-N per kg N excreted	Selected value from Table 5.5	Secondary data from IPCC, 2019; see Table 5.5	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining EF based on system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
$Frac_{leach_{MS}}^T$	Fraction of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in manure management system S	Fraction	Selected value from Table 5.7	Secondary data from IPCC, 2019; see Table 5.7	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining selection based on system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

EF_5	Emission factor for N ₂ O emissions from nitrogen leaching and runoff	kg N ₂ O-N per kg N leached and runoff	Selected value from Table 5.8	Secondary data from IPCC, 2019; see Table 5.8	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining selection based on system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
$Frac_{gas_{MS}}^T$	Fraction of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x in the manure management system S	Fraction	Selected value from Table 5.7	Secondary data from IPCC, 2019; see Table 5.7	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining selection based on system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.

EF_4	Emission factor for N ₂ O emissions from nitrogen volatilization	kg N ₂ O-N per kg NH ₃ -N + NO _x -N volatilized	Selected value from Table 5.8	Secondary data from IPCC, 2019; see Table 5.8	Every reporting period	Values are reviewed during QA/QC process & audited at verification	Producer: inputting necessary values and ensuring they are correct / Elanco: Responsible for determining selection based on system / Verifier: corroborating values against documentation	Data and optionally supporting documentation are stored in UpLook.
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7 Reporting

Project developers must provide the following documentation each reporting period to generate credits from this protocol:

1. Name and address of the project developer
2. List of all of the operations included in the project including the owner/operator contact information and address of the operation
3. Regulatory compliance documentation and attestation
4. Monitoring plan
5. Monitoring report with all the data used in the calculations for Section 5 of the protocol
6. Monitoring report must include the intended use and user of the monitoring report.

7.1 Record Keeping

For purposes of third-party verification and historical documentation, project developers must keep all information listed in this protocol for a period of 10 years after the information is generated. The information the project developer should retain includes:

1. All data inputs for the calculation of the project emission reductions as well as the results of emission reduction calculations
2. Copies of all permits, Notices of Violations (NOVs), and any relevant administrative or legal orders dating back at least 3 years prior to the project start date
3. All verification records and results
4. All maintenance records relevant to the monitoring equipment

8 Verification

Verification bodies will contract directly with Athian for all validation and verification engagements.

Projects verified under this protocol will meet, at minimum, the auditing standard of limited assurance and adhere to 14064-3. The verification body must provide a factual statement expressing the outcome of the verification.

Issues identified during verification must be classified by verification bodies as either material (significant) or immaterial (insignificant). To be verified successfully, all reported emissions reductions must be free of material misstatements.

All projects developed under this protocol must achieve >95 percent level of accuracy. This means that the project's calculated emission reductions must be less than 5 percent different than those calculated by the verifier.

8.1 Verification Body Requirements

To conduct verification under this protocol, all Validation and Verification Bodies (VVB) must meet the following criteria:

1. Accreditation under International Organization for Standardization (ISO) 14065: 2013 with conformance to all accreditation requirements under ISO 14065, ISO 14064-3: 2006, IAF MD 6: 2014 and all other accreditation requirements, or Acceptance in the American National Standards Institute (ANSI) accreditation program, having filed a full application for ISO 14065: 2020
2. Demonstrated/documented subject matter expertise in the on-farm operations related to an approved protocol (e.g., Dairy Operations; Feed Lot Operations)
3. Demonstrated/documented experience in a particular region or state where the verification will occur
4. Monitoring conducted in accordance with the requirements of the relevant protocol
5. Monitoring conducted in a manner that allows for a complete and transparent quantification of GHG reductions

8.2 Conflict of Interest

When conducting verification under this protocol verifiers must be seen as credible, independent, and transparent. To meet this requirement, a conflict of interest (COI) determination must be made prior to starting any verification activities. A COI occurs in any situation that compromises the verifier's ability to perform an independent verification. Every verifier must provide information about its organizational relationships, internal structures, and management systems for identifying potential COIs. Verifiers must evaluate any potential conflicting services it has provided to the project developer, including any advice or consulting provided outside of the verification process.

8.3 Verification Process

To verify the project, the verifier must develop a risk-based verification plan that considers the size and complexity of the project and the relevant sector, technology, and processes. The verifier must follow the following process:

1. Complete a COI evaluation. If there is a potential COI, the verifier is not allowed to conduct the verification.
2. Prepare a verification plan that includes, at a minimum:
 - a. A list of people from the VVB involved in the verification
 - b. A list of the location and dates of any on-site visits that will be conducted
 - c. The types of data and documents that will be reviewed by the verifier
 - d. A list of the people who are expected to be interviewed as a part of the verification
3. Conduct a kick-off meeting with all parties to lay out the timeline and process of the verification.
4. Conduct, at minimum, one annual on-site visit to confirm practice implementation

5. Undertake a desk review of the data from the project.
6. Prepare a verification report that includes:
 - a. A verification statement documenting the outcome of the assessment (reduction results) and if there was any material discrepancy noted
 - b. Key details about the project including: producer and farm operation identification, verifying body and lead verifier contact information, protocol information, and intervention information
 - c. A description of the protocol, the objectives and criteria used to arrive at the final result, the scope of the project, the level of assurance associated with the project, and any details about the implementation of the practices observed
 - d. Detail about the verification process used to complete the assessment including approach and methods and also noting any conflict of interest
 - e. Verification findings including confirmation of producer eligibility, adherence to the criteria established in the protocol, the verified emissions quantification values, and the final written opinion of the verifier(s)
 - f. An issue log capturing any issues identified during the verification and their classification as either material (significant) or immaterial (insignificant)
 - g. A representation of all data / documents used in the process of verification

9 References

- Adom, F. et al. (2012) Regional carbon footprint analysis of dairy feeds for milk production in the USA. *Int. J. Life Cycle Assess.* 17:520–534. <http://doi.org/10.1007/s11367-012-0386-y>
- Adom, F. et al. (2013). Carbon footprint analysis of dairy feed from a mill in Michigan, USA. *Int. Dairy J.* 31(1):S21–S28. <https://doi.org/10.1016/j.idairyj.2012.09.008>
- Appuhamy, J.A.S.R.N. et al. (2013) Anti-methanogenic effects of monensin in dairy and beef cattle: A meta-analysis. *J. Dairy Sci.* 96:5161–5173. <http://dx.doi.org/10.3168/jds.2012-5923>
- Dämmgen U. et al. (2011) Reassessment of the calculation procedure for the volatile solids excretion rates of cattle and pigs in the {Austrian}, {Danish} and {German} agricultural emission inventories. *Landbauforschung Volkenrode* 61:115–126.
- Department for Environment, Food and Rural Affairs (2012). Guidelines to Defra / DECC's GHG Conversion Factors for Company Reporting: Methodology Paper for Emission Factors. July 2012. PB 13792. <https://assets.publishing.service.gov.uk/media/5a74a089ed915d0e8bf19d74/pb13792-emission-factor-methodology-paper-120706.pdf> Accessed June 21st, 2024.
- Dijkstra, J., Bannink, A., France, J., Kebreab, E., & van Gastelen, S. (2018). Short communication: Antimethanogenic effects of 3-nitrooxypropanol depend on supplementation dose, dietary fiber content, and cattle type. *Journal of Dairy Science*, 101(10), 9041–9047. <https://doi.org/10.3168/jds.2018-14456>
- Duffield, T.F. et al. (2008) A Meta-Analysis of the Impact of Monensin in Lactating Dairy Cattle. Part 2. Production Effects. *J. Dairy Sci.* 91:1347–1360. <https://doi.org/10.3168/jds.2007-0608>

Duin, E. C., Wagner, T., Shima, S., Prakash, D., Cronin, B., Yáñez-Ruiz, D. R., Duval, S., Rübels, R., Stemmler, R. T., Thauer, R. K., & Kindermann, M. (2016). Mode of action uncovered for the specific reduction of methane emissions from ruminants by the small molecule 3-nitrooxypropanol. *Proceedings of the National Academy of Sciences of the United States of America*, 113(22), 6172–6177. <https://doi.org/10.1073/pnas.1600298113>

Environmental Protection Agency (2024) GHG Emissions Factors Hub data. <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>. Accessed June 21st, 2024.

Environmental Protection Agency (2023) Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2021. U.S. Environmental Protection Agency, EPA 430-R-23-002. <https://www.epa.gov/ghgemissions/inventory-us-greenhousegas-emissions-and-sinks-1990-2021>.

Food and Agriculture Organization of the United Nations (2017). Global database of GHG emissions related to feed crops: A life cycle inventor <https://www.fao.org/partnerships/leap/database/ghg-crops/en/>

FDA (U.S. Food and Drug Administration) (2004) New Animal Drugs for Use in Animal Feeds; Monensin, 69 Fed. Reg. 6557 (February 11, 2004) (codified at 21 C.F.R. pt. 558)

Gadberry, S. et al. (2022) Meta-analysis of the effects of monensin on performance of beef replacement heifers and beef cows. *Trans. Anim. Sci.* 6:1–9. <https://doi.org/10.1093/tas/txac086>

Gerber, P. J., Steinfeld, H., Henderson, B., Mottet, A., Opio, C., Dijkman, J., Falcucci, A., & Tempio, G. (2013). Tackling Climate Change Through Livestock: A Global Assessment of Emissions and Mitigation Opportunities. In Food and Agriculture Organization of the United Nations.

GitarSKIY, M. L. (2019). The refinement to the 2006 IPCC guidelines for National Greenhouse Gas Inventories. *Fundamental and Applied Climatology*, 2, 5–13. <https://doi.org/10.21513/0207-2564-2019-2-05-13>

Hristov, A. N., Melgar, A., Wasson, D., & Arndt, C. (2022). Symposium review: Effective nutritional strategies to mitigate enteric methane in dairy cattle. *Journal of Dairy Science*, 105(10), 8543–8557. <https://doi.org/10.3168/jds.2021-21398>

Harris, I., Jones, P. D., Osborn, T. J. & Lister, D. H. (2014) Updated High-resolution Grids of Monthly Climatic Observations – the CRU TS3.10 Dataset. *International Journal of Climatology* 34(3): 623–642.

IPCC, 2019. “Chapter 10: Emissions from Livestock and Manure Management.” In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Volume 4. Agriculture, Forestry and Other Land Use. IPCC. 2019. Buendia, E. et al., Eds. <https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html>

IPCC, 2021. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2391 pp. <https://doi.org/10.1017/9781009157896>

Jayanegara, A., Sarwono, K. A., Kondo, M., Matsui, H., Ridla, M., Laconi, E. B., & Nahrowi. (2018). Use of 3-nitrooxypropanol as feed additive for mitigating enteric methane emissions from ruminants: a meta-analysis. *Italian Journal of Animal Science*, 17(3), 650–656.
<https://doi.org/10.1080/1828051X.2017.1404945>

Kebreab, E., Bannink, A., Pressman, E. M., Walker, N., Karagiannis, A., van Gastelen, S., & Dijkstra, J. (2023). A meta-analysis of effects of 3-nitrooxypropanol on methane production, yield, and intensity in dairy cattle. *Journal of Dairy Science*, 106(2), 927–936.
<https://doi.org/10.3168/jds.2022-22211>

Kim, H., Lee, H. G., Baek, Y. C., Lee, S., & Seo, J. (2020). The effects of dietary supplementation with 3-nitrooxypropanol on enteric methane emissions, rumen fermentation, and production performance in ruminants: A meta-analysis. *Journal of Animal Science and Technology*, 62(1), 31–42. <https://doi.org/10.5187/JAST.2020.62.1.31>

Martins, L. F., Maigaard, M., Johansen, M., Lund, P., Ma, X., Niu, M., & Hristov, A. N. (2025). Lactational performance effects of 3-nitrooxypropanol supplementation to dairy cows: A meta-regression. *Journal of Dairy Science*, 108(2), 1538–1553.
<https://doi.org/10.3168/jds.2024-25653>

National Academies of Sciences, Engineering, and Medicine (NASEM). 2023. Nutrient Requirements of Dairy Cattle: Eight Revised Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25806>

Nkemka, V., Beauchemin, K., & Hao, X. (2019). Treatment of feces from beef cattle fed the enteric methane inhibitor 3-nitrooxypropanol. *Water Science and Technology*, 80(3), 437–447. <https://doi.org/10.2166/wst.2019.302>

Odongo, N. E., Bagg, R., Vessie, G., Dick, P., Or-Rashid, M. M., Hook, S. E., Gray, J. T., Kebreab, E., France, J., & McBride, B. W. (2007). Long-term effects of feeding monensin on methane production in lactating dairy cows. *Journal of Dairy Science*, 90(4), 1781–1788.
<https://doi.org/10.3168/jds.2006-708>

Owens, J., Thomas, B., Stoeckli, J., Beauchemin, K., McAllister, T., Larney, F., & Hao, X. (2020). Greenhouse gas and ammonia emissions from stored manure from beef cattle supplemented 3-nitrooxypropanol and monensin to reduce enteric methane emissions. *Scientific Reports*, 10(1), 1–14. <https://doi.org/10.1038/s41598-020-75236-w>

Owens, J., Hao, X., Thomas, B. W., Stoeckli, J., Soden, C., Acharya, S., & Lupwayi, N. (2021). Effects of 3-nitrooxypropanol manure fertilizer on soil health and hydraulic properties. *Journal of Environmental Quality*, 50(6), 1452–1463. <https://doi.org/10.1002/jeq2.20276>

Pupo, M. R., Ferraretto, L. F., & Nicholson, C. F. (2025). Effects of feeding 3-nitrooxypropanol for methane emissions reduction on income over feed costs in the United States. *Journal of Dairy Science*. <https://doi.org/10.3168/jds.2024-25502>

Romero-Pérez, A., Okine, E. K., Guan, L. L., Duval, S. M., Kindermann, M., & Beauchemin, K. A. (2016). Effects of 3-nitrooxypropanol and monensin on methane production using a forage-based diet in Rusitec fermenters. *Animal Feed Science and Technology*, 220, 67–72.
<https://doi.org/10.1016/j.anifeedsci.2016.07.013>

Rotz, A. et al. (2021) Environmental assessment of United States dairy farms. *Journal of Cleaner Production*. 315:128153. <https://doi.org/10.1016/j.jclepro.2021.128153>

Uddin, M. E., Tricarico, J. M., & Kebreab, E. (2022). Impact of nitrate and 3-nitrooxypropanol on the carbon footprints of milk from cattle produced in confined-feeding systems across regions in the United States: A life cycle analysis. *Journal of Dairy Science*, 105(6), 5074–5083. <https://doi.org/10.3168/jds.2021-20988>

van Gastelen, S., Burgers, E. E. A., Dijkstra, J., de Mol, R., Muizelaar, W., Walker, N., & Bannink, A. (2024). Long-term effects of 3-nitrooxypropanol on methane emission and milk production characteristics in Holstein-Friesian dairy cows. *Journal of Dairy Science*, 107(8), 5556–5573. <https://doi.org/10.3168/jds.2023-24198>

Vyas, D., McGinn, S. M., Duval, S. M., Kindermann, M., & Beauchemin, K. A. (2016). Effects of sustained reduction of enteric methane emissions with dietary supplementation of 3-nitrooxypropanol on growth performance of growing and finishing beef cattle. *Journal of Animal Science*, 94(5), 2024–2034. <https://doi.org/10.2527/jas.2015-0268>